



Artificial Intelligence for Elderly Fall Detection: State-of-the-art Methods, Applications and Challenges

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Abstract

Accidental falls have emerged as a major public health concern, especially among individuals aged 65 and older, due to their high incidence and severe consequences. Without timely intervention, such falls can result in fractures, traumatic brain injuries, and long-term complications. As a result, considerable research has focused on developing automated fall detection systems that integrate intelligent algorithms with sensor-based data acquisition to enable rapid response and medical assistance. This study follows the PRISMA framework to conduct a systematic literature review. A comprehensive search was performed across major databases including PubMed, Google Scholar, Scopus, and IEEE Xplore using fall detection-related keywords. From an initial pool of 596 articles, duplicates were removed and strict inclusion/exclusion criteria were applied, resulting in 182 relevant articles for in-depth analysis. This review examines a wide range of Artificial Intelligence (AI) and Machine Learning (ML) approaches applied to fall detection using diverse sensor modalities, including wearable, vision-based, ambient, and multimodal systems. Additionally, it summarizes the publicly available datasets and sensor configurations used in existing studies. This broad perspective is essential to compare trade-offs across sensing modalities and support effective system development for real-world deployment. This review provides a comparative overview of AI/ML-based fall detection approaches, highlighting differences in accuracy, sensitivity, and dataset usage as reported in existing literature. In the end, many open research challenges in using AI and ML to detect falls are outlined, along with potential future perspectives.

Keywords Wearable sensor · Vision · Ambient · Sensor fusion · Machine learning · Deep learning

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Introduction

The world is beginning to observe a prolonged change in demographic age distribution, owing to the increasing average lifespan and decreasing birth rate trends. People live longer lives now than before. The percentage of older people in the overall population is growing tremendously. In 2020, 727 million people worldwide were reported to be 65 or older [1]. The global elderly population is projected to almost double over the next three decades, reaching over 1.5 billion by 2050 [1]. Nearly every region of the world will experience an increase in the ageing population by 2050. The percentage of people aged 65 and up is expected to rise from 9.3% to 16.0% within thirty years [2]. Ageing is often associated with deteriorating functionalities, such as physiological, visual, and cognitive impairments, which increase the likelihood of falling. Around 28–35 per cent of elderly individuals aged 65 or above every year are estimated to fall [3]. As age and frailty increase, the frequency of falls significantly increases. An estimated 646,000 fatal falls occur annually, making it the second leading cause of mortality by accidental injuries after road accidents [4]. Nearly 37.3 million falls each year require to receive medical treatment [5].

More than 17 million disability-adjusted life years (DALYs) have been lost as a result of these falls [4]. Fall-related deaths account for almost 80% of deaths in low and middle-income countries, with 60% of these deaths happening in Southeast and Western Pacific Asia [6]. Senior citizens over 70 years who fall each year and suffer serious or minor medical injuries are estimated to be between 32 to 42 per cent [7]. Fall is one of the unexpected occurrences in regular activities, impacting individual health.

Fall events demonstrate loss of voluntary body control and drastic shifts in body acceleration. Additionally, fall occurrences create monetary and mental pressure on the caregivers and decrease life quality. Early detection of falls will help to reduce severe effects and support healthcare [8–12] sector. Fall recognition and prevention are two strategies for responding earlier to elderly fall events. Researchers studied these strategies, particularly various fall detection methods explored and discovered over the past two decades. Different Artificial Intelligence and Machine Learning [13–23] algorithms are examined alongside the utilisation of multiple kinds of sensors for fall detection (see Fig. 1).

A significant number of literature reviews are performed on fall detection and prediction systems. Most review works cover specified categories and technologies implemented for fall detection and prevention. Pannurat et al. [24] reviewed existing elderly fall detection schemes based on wearable and ambient sensors, prioritising wearable sensor-based systems. The authors also analysed commercially available wearable fall-detecting devices and studied the most useful statistical features implemented. A systematic literature review has been carried out [25] to investigate wearable elderly fall detection systems. These were differentiated into three groups: fall risk assessment, fall type identification, and fall events detection. The authors studied previously selected research questions among 29 selected peer-reviewed journals and analysed the cutting-edge practice in the existing systems. Limitations and research gaps of the studied articles are discussed, along with scopes and directions for future research. Chaudhuri et al. [26] presented a systematic literature review on fall detection devices. Their acceptance

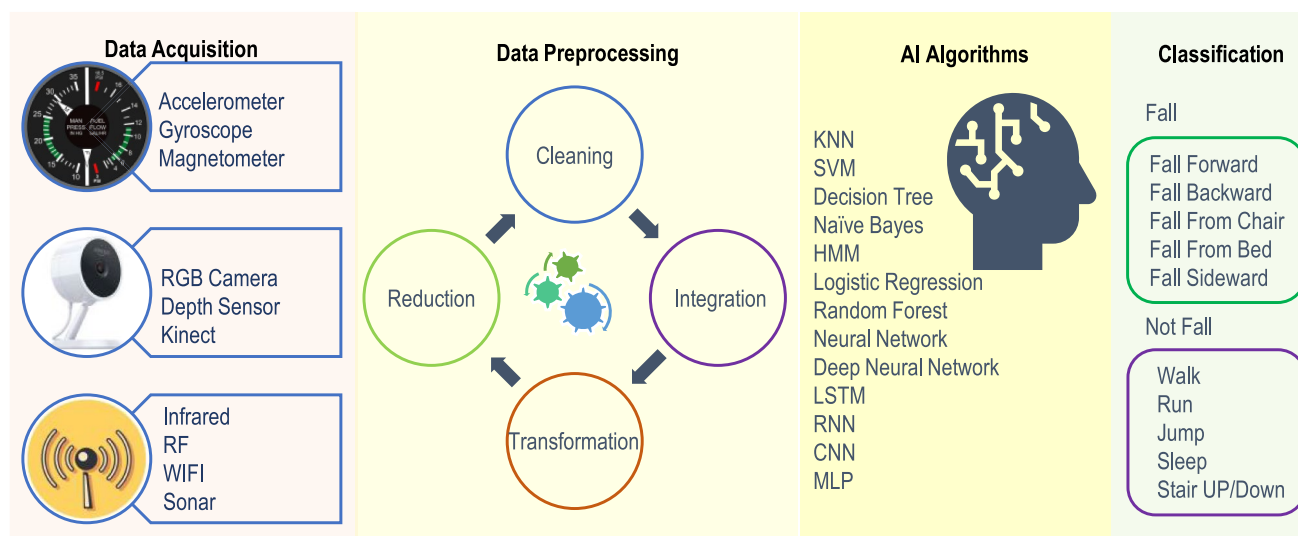


Fig. 1 A general architecture of Artificial Intelligence (AI) algorithm-based Fall Detection System (FDS). Different types of sensors (Wearable/Vision/Ambient) are utilized to collect human movement data. Sensor data are then preprocessed and fed into an AI-based model to detect falls

towards the target elderly was assessed by investing in the implementation and designing procedures of several state-of-the-art fall detection devices. The authors divided the studied works into two categories: based on evaluation criteria and based on device wearability (wearable and non-wearable sensor-based systems). Cippitelli et al. [27] presented a survey on the state of the fall detection systems presented in the literature based on RADAR sensors and RGB-depth (RGBD) sensors. Different RADAR and RGBD sensors were explored in this study, along with several feature extraction techniques, algorithms and evaluation criteria used in selected existing fall detection systems. Several open research challenges were identified, and directions to the field's future research trends were provided in the paper. Khan et al. [28] developed a review on human fall detection approaches from the point of availability of fall dataset perspective. They divided the existing systems into two categories: enough fall data and insufficient fall training data, completely independent of feature extraction methods and the type of sensors used in the systems. Due to less real fall training data, the authors identified the fall detection system as an anomaly detection problem that belongs to the insufficient data category. Several ML algorithms were discussed, which work well in fall-related data. Finally, the authors suggest future research directions for efficiently identifying fall events. Ramachandran et al. [29] surveyed the application of ML algorithms in wearable fall detection systems with an overview of different sensor-based fall detection systems such as vision sensors and environment and wearable sensors. Several research works were investigated to determine the implemented wearable sensors and sensor placement, ML algorithms and their performances. The authors also classified wearable sensor-based systems into threshold-based and ML-based fall detection systems. The survey concluded by discussing the effect of biological and personal profiles on detecting fall events. Nizam et al. [30] conducted a systematic review of fall detection methods based on human activity classification and divided the fall detection systems into wearable, ambient and vision-based categories. A detailed discussion on each category was presented based on different sensor technologies. The authors added a brief review of depth sensor-based fall detection systems and compared their performances. A brief summary of recently published reviews on fall detection is shown in Table 1 where limitations of existing reviews are included.

Although the studies mentioned earlier cover the current state-of-the-art fall detection approaches, most surveys only focus on wearable or vision-based fall detection systems. Most of the systematic reviews are focused on a specific part of fall detection systems, omitting the overall fall detection

approaches. To address this research gap, we present a literature study of different fall detection systems following the PRISMA research protocol. In this paper, a systematic study is performed that provides an overall review of fall-related systems categorically. Types of sensors and AI algorithms used for existing Fall detection systems are analysed alongside the success rate of the proposed systems.

If you are new to the field of fall detection, this review serves as a structured and practical entry point to the domain. We present a comprehensive taxonomy of fall detection systems, categorized by sensor modality—wearable, vision-based, ambient, and multimodal—and by algorithmic strategy, including threshold-based and machine learning approaches. This classification framework helps readers grasp the landscape of existing approaches, their comparative advantages, and limitations. In addition, detailed comparisons of datasets, sensor types, and performance metrics are provided to assist researchers in selecting appropriate benchmarks and identifying research gaps where new contributions can be impactful. Unlike prior surveys that narrowly focus on one or two modalities or specific algorithm types, this review takes a holistic approach by offering a structured taxonomy across multiple sensing paradigms and algorithmic layers. This enables a critical analysis of cross-modality trade-offs and supports informed decision-making for future FDS development.

The main contributions of the paper are as follows:

1. This study covers existing elderly fall detection research works published in journals, book chapters, and conference proceedings between 2006 and 2025.
2. The fall detection approaches are categorised into four different types based on implemented sensor types—Wearable sensors, Vision sensors, Ambient sensors, and Multimodal sensors and subdivided each approach based on the implemented algorithms, including threshold-based algorithms and different AI algorithms.
3. Different types of data acquisition sensors that have been used in various fall detection works are discussed.
4. It discusses available open-access datasets.
5. The performance of different fall detection approaches is critically analysed based on their used methods.
6. Finally, the challenges, limitations of elderly fall detection, and specific research directions for future enthusiastic researchers are given in this paper.

The rest of the article is organised as follows. The research methodology for this review is described in the “Survey Methodology” section. Different types of sensors are described in the “Sensors Used in Different Fall Detection Studies” section. Sections such as “Wearable Sensor-based

Table 1 Summary and limitations of recently published review articles on fall detection approaches

Ref.	Year	Modality	Model	Sys-tematic Review	Publication year range of studied articles	Number of studies included	Limitations
[24]	2014	Wearable, Vision, Ambient, Multimodal	Threshold, ML	No	Up to 2014	60	No deep dive into deep learning methods; no benchmark/meta-analysis; lacks real-world elderly data; wearable fusion underdeveloped
[25]	2019	Wearables	ML	Yes	2002 to 2019	29	Limited databases searched
[26]	2014	Wearable, Vision, Ambient, Multimodal	Threshold, ML	Yes	Up to 2013	113	Manual screening bias; lack of real-world validation; inconsistent evaluation criteria; few long-term studies
[27]	2017	Vision, Ambient	Threshold, ML, DL	No	2011 to 2017	40	No quantitative meta-analysis; narrow modality focus
[28]	2017	Wearable, Vision, Ambient, Multimodal	Threshold, ML	No	Up to 2016	Not mentioned	No performance meta-analysis; does not compare sensor fusion systems
[29]	2020	Wearable	Threshold, ML, DL	No	2005 to 2019	Not mentioned	No meta-analysis or statistical comparison of model performances
[30]	2020	Wearable, Vision, Ambient, Multimodal	Threshold, ML, DL	Yes	2005 to 2019	40	Fusion techniques are underexplored and unsynchronized
[31]	2021	Wearable Sensor, Vision, WiFI, Radar	ML	No	Not mentioned	Not mentioned	No dataset description, No recommendations for future research, A very limited number of studies included
[32]	2021	Vision	ANN, ML	Yes	2015 to 2020	81	No recommendation, Only confined to Vision-based FDS
[33]	2021	Wearable and Now-wearable	ML	Yes	2010 to 2020	33	Not covered Deep Learning and Threshold-based models
[34]	2022	Wearables	ML, DL, TBA	Yes	2017 to 2020	85	Not covered Vision/ambient/Multimodal based FDS
[35]	2021	Wearables	Models were not concerned	Yes	2014 to 2019	Not mentioned	Only limited to Single/Multi-sensor based FDS
[36]	2022	Vision	DL	Yes	2014 to 2020	74	Only limited to Vision-based deep learning approaches
[37]	2021	Not mentioned	Not mentioned	No	2015 to 2020	54	Only focused on technologies; not methods
[38]	2022	Sensor	ML	No	2007 to 2020	13	A very limited number of studies
[39]	2021	Vision, Ambient, Wearable	Not mentioned	Yes	2015 to 2019	11	Limited to journal articles, Only 11 articles, No discussion about dataset
[40]	2022	PIR	CNN/LSTM	No	2015 to 2020	15	Investigated the metrological qualities of PIR sensors, Not focused on Fall detection approaches
[41]	2023	Vision, Wearable, Non-Wearable	ML, DL	Yes	2017 to 2023	26	Not covered beyond 2023; only focused on ML and DL; did not include multimodal approaches
[42]	2024	Radar	ML, DL	Yes	2000 to 2024	74	Limited to Radar-based fall detection systems only
[43]	2024	Radar	DL	No	NA	NA	Only limited to Radar-based fall detection systems; lacks generalizability across modalities

Fall Detection Approaches”, “Vision Sensor-based Fall Detection Approaches”, “Ambient Sensor-based Fall Detection Approaches”, and “Multimodal Sensor-based Fall Detection Approaches” demonstrate the fall detection approaches based on wearable, vision, ambient, and multimodal sensors, respectively. The selected articles’ outcomes are critically analysed in the “Performance Analysis” section. Existing Challenges and Future Works” outlines the existing challenges in fall detection and provides some recommendations. Lastly, “Conclusion” wraps up the entire work and concludes the study.

Survey Methodology

Based on previous studies, this review describes the methods of detecting falls and how they can help researchers develop new AI-based approaches for fall detection. We have used four scholarly databases (Google Scholar, Pubmed, Scopus, and IEEEExplore) to download fall detection research papers that have been published between 2006 and 2025.

The previous literature on fall detection was comprehensively analysed and treated without bias by searching

databases using search terms directly related to fall detection. As mentioned earlier, the search terms for the databases are given in Table 2. We have found 596 articles from four databases using the selected search terms. Then, we removed the duplicate articles from the download articles. Finally, we have selected 182 articles for this review based on some inclusion and exclusion criteria. The detail of the survey methodology is shown in Fig. 2.

The rationale for this review is that research papers on fall detection have been published incredibly in recent years, thus requiring them to be assessed, classified, and analysed. This study aims to determine whether any AI methodologies have been applied for fall detection based on publicly available datasets. This research is intended for the new researchers who will analyse and detect falling events. Also, interdisciplinary researchers from computer science, biology, and medicine.

Sensors Used in Different Fall Detection Studies

Fall detection systems have achieved substantial advances due to the use of advanced sensors. Sensor technology has progressed remarkably in recent years, which provides humans with the ability to collect more precise and convenient information from their surroundings at any given point

in time. We have categorised fall detection studies based on different types of sensors. Table 3 presents the different types of sensors and their methods, ranges, advantages, and potential drawbacks.

The Accelerometer sensor is the most used one in wearable sensors. It is used to collect acceleration information due to falls. Gyroscope sensors are utilised to measure a tracked person's rotation and angular velocity. Magnetometer sensors are also incorporated in many studies to detect falls. Furthermore, various sensors, such as ECG, sEMG, FSR, etc., are used in wearable Fall detection systems (FDS). Apart from wearable sensors, vision sensor-based FDS has become popular with computational technology advancement in recent years. Depth cameras and RGB cameras are the most utilised sensing equipment in vision-based FDS. Alongside wearable and vision sensors, ambient sensor-based FDS has exhibited improved performance and earned high acceptance towards the elderly. Ambient sensors like Radio-Frequency Identification (RFID), Acoustic sensors, Radar sensors, Pressure sensors, etc., improve the FDSs significantly.

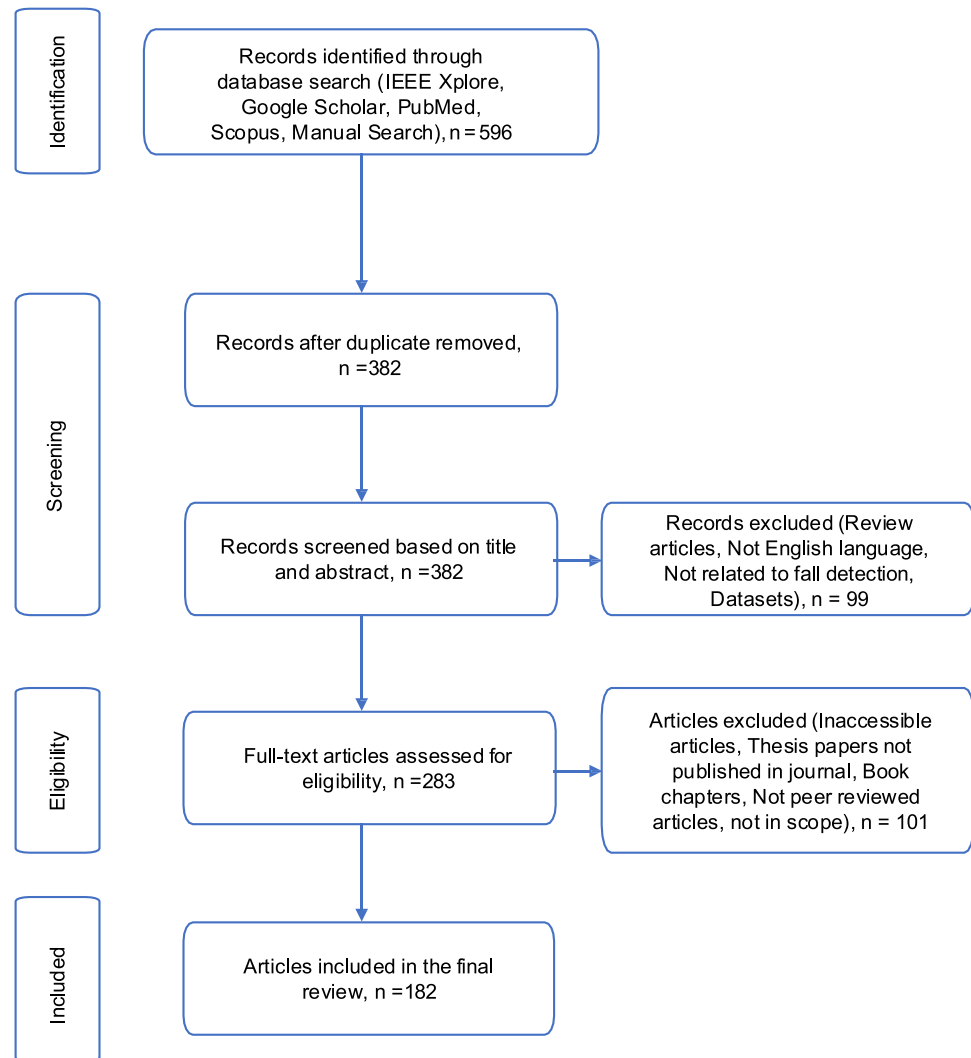
Wearable Sensor-based Fall Detection Approaches

Research on wearable sensor-based fall detection has gained popularity in the research community since 2006. Researchers are attempting to detect falls utilising ML and threshold-based algorithms. Figure 3 represents a general architecture of the wearable FDS. This section aims to provide a better insight into wearable sensor-based FDSs that use ML and threshold-based algorithms. Table 4 provides a comprehensive overview of the sensor types, placement locations, and extracted features used across a wide range of wearable systems. It highlights the impact of sensor modality (e.g., accelerometer, gyroscope, pressure sensors) and position (e.g., waist, wrist, foot) on detection accuracy and reliability. Additionally, the feature sets vary from handcrafted statistical descriptors (e.g., SMV, variance, entropy) to learned temporal representations, reflecting the algorithmic diversity of these approaches. This diversity emphasizes the need to align sensor configuration and feature design with the target deployment scenario, especially for achieving real-time and resource-efficient detection. We have collected publicly available datasets used in the selected articles for this work. A comparative summary of different publicly available datasets is presented in Table 5. The following subsections discuss wearable sensor-based fall detection systems and their working process based on algorithms and the sensor data they used.

Table 2 Databases and queries for literature search

Database	Query
Pubmed	(((((("Video"[Title/Abstract] OR "Vision"[Title/Abstract]) OR "Image"[Title/Abstract]) OR "Camera"[Title/Abstract]) OR "Wearable"[Title/Abstract]) OR "Ambient"[Title/Abstract]) OR "Inertial"[Title/Abstract]) OR "body worn"[Title/Abstract]) AND "fall detection"[Title/Abstract]) AND ("journal article"[Publication Type] AND "aged"[MeSH Terms])
IEEE Xplore	(((((("All Metadata": "Video") OR "All Metadata": "Vision") OR "All Metadata": "Camera") OR "All Metadata": "Image") OR "All Metadata": "Not Wearable") OR "All Metadata": "Wearable") OR "All Metadata": "Inertial") OR "All Metadata": "Body Worn") OR "All Metadata": "Ambient") AND "All Metadata": "Fall Detection")
Google Scholar	allintitle: Video OR Vision OR Camera OR Image OR Wearable OR Ambient OR "Body Worn" OR Inertial OR Unobtrusive AND ("Elderly Fall" OR "Fall Detection" OR "Fall Detector")
Scopus	(TITLE (fall AND detection) AND TITLE (audio) OR TITLE (vision) OR TITLE (ambient) OR TITLE (wearable) OR TITLE (multimodal) OR TITLE (video) OR TITLE (camera))

Fig. 2 The systematic selection procedure of the studied articles based on PRISMA search strategies. At first, a list of 596 articles was obtained through a search string from Scopus, Google Scholar, IEEE Xplore and PubMed databases. Finally, 182 articles have been selected through the PRISMA steps



Threshold-Based Approaches

Threshold-based approaches have been extensively utilized in wearable Fall Detection Systems (FDS), particularly leveraging accelerometer sensors. Many studies focus on integrating accelerometer data with additional sensors or communication technologies to enhance detection accuracy, power efficiency, and real-time capabilities.

Accelerometer-Only Systems

Numerous studies rely solely on accelerometer data to detect falls by setting predefined thresholds. For instance, Qian et al. [44] developed a system with a multilevel threshold algorithm, achieving high accuracy and scalability through real-time processing with MEMS sensors and Narrow Band IoT. Similarly, Gannapathy et al. [45] utilized waist-mounted 3D accelerometers and Zigbee transceivers to notify contacts of detected falls. Sucerquia et al. [46]

implemented a low-computing-cost threshold algorithm using zero-crossing and Kalman filtering techniques, validated on the SisFall dataset. These systems emphasize simplicity and robustness in fall detection.

Smartphone-Based FDS

Integrating accelerometers with smartphones has also been explored. Basili et al. [47] and Yi et al. [48] implemented systems combining smartphone sensors with external accelerometers, focusing on human posture analysis and real-time fall alerts. Enhanced by additional sensors like ECG [49], these systems expanded functionality while retaining threshold-based simplicity.

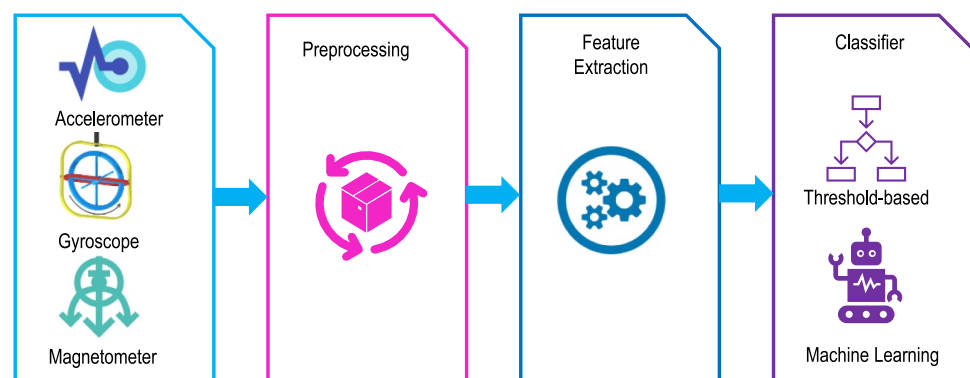
Multi-Sensor Systems

To improve detection accuracy, accelerometers are often combined with other sensors. Combining a barometer with

Table 3 Description, range, advantages, and disadvantages of different sensor categories

Category	Sensor	Description	Detection Range	Cost	Ease of Setup	Power Consumption	Durability	Security	Precision
Wearable	Accelerometer	Measures acceleration in three axes.	$\pm 16g$	Low	Easy	Low	Moderate	High	High
	Gyroscope	Measures rotation and angular velocity.	± 1000 °/s	Moderate	Easy	Low	High	High	High
	Magnetometer	Measures magnetic field strength and spatial orientation.	± 1900 μT	Low	Easy	Low	High	High	Low
	ECG	Captures electrical activity of the heart.	$\pm 1.5mV$	Low	Easy	Low	High	Moderate	High
	sEMG	Records electrical activity of skeletal muscles.	$\pm 1.5mV$	High	Easy	Low	High	High	Low
	FSR	Measures pressure and force via resistance changes.	Up to 150 kg	Low	Easy	Low	High	High	Low
	Barometer	Measures atmospheric pressure for altitude estimation.	300 to 1100 hPa	Low	Easy	Low	High	High	High
Vision	Depth Camera	Measures distance using the speed of light.	Up to 10 m	High	Moderate	High	High	High	High
	RGB Camera	Captures visible light images in red, green, and blue channels.	Up to 10 m	Moderate	Moderate	High	High	Moderate	High
Ambient	RFID	Detects tags using radio frequency signals.	Up to 10 m	Moderate	Easy	Low	High	High	Moderate
	WiFi	Uses radio frequency for wireless communication.	Up to 50 m	Low	Easy	Moderate	High	High	High
	Acoustic	Measures sound levels via acoustic waves.	20 Hz to 20 kHz	Low	Easy	Low to High	High	High	Low
	Pressure	Measures force applied on a surface.	15 to 115 kPa	Low	Easy	Low	High	High	High
	Radar	Detects objects using radio waves.	Up to 100 m	High	Difficult	High	High	High	High
	Infrared	Detects warm objects using infrared radiation.	10 to 30 m	Low	Easy	Low	High	High	High

Fig. 3 Wearable sensor-based fall detection block diagram where body-worn sensors such as accelerometer, gyroscope, magnetometer, and heart rate sensor can be used. Sensor-recorded data can be passed through preprocessing, feature extraction, and classification steps to detect falls



an accelerometer [50, 51] improves fall detection by providing altitude information to distinguish between falls and other movements, such as lying down or sitting. Accelerometer-gyroscope systems [52, 53] leverage angular velocity for more precise fall classification. Systems like those proposed by Sabatini et al. [54] even detect both pre- and

post-fall impacts using barometric altimeters. Vyas et al. [55] developed a non-intrusive wearable fall detection system by embedding piezoresistive textile pressure sensors and an accelerometer in socks. They have employed a threshold-based dual-check algorithm to distinguish falls from normal activities.

Table 4 Sensor positions and extracted features for wearable fall detection approaches

Ref.	Year	Sensors	Sens. Pos.	Features
[55]	2025	Pres, Acc	Sock	Pressure readings, resultant acceleration vector, smoothed acceleration
[56]	2025	Acc, Gyr	Waist / Lower back	Raw time-series data, auto-feature extraction via CNN-BiLSTM
[57]	2024	Acc	Leg	FFT features, filtered signal, statistical and temporal data sequences
[58]	2024	Pres	Foot insole	Time-series stress video from plantar pressure, interpolated 10×10 maps
[59]	2022	Acc	Waist	SMV, spatial orientation, filtered angle change
[60]	2022	Acc, Gyr	Wrist, Waist, Ankle	Raw accelerometer and gyroscope data
[61]	2023	Acc, Gyr	Wrist	88 statistical and spectral global features
[62]	2023	Acc, Gyr	Shank, Pelvis, Upper/Lower Arms, Forearm, Thoracic, Foot, Hip, Hand, Head	Raw time-series and convolutional learned features
[63]	2021	Acc, Gyr	Waist	Time-series data, spectral features
[64]	2024	Acc, Gyr	Waist, Thigh	Spatial and Temporal Features
[65]	2023	Acc, Gyr	NA	Acceleration, Angular Velocity
[44]	2022	Acc, Gyr	Wrist	SMV
[66]	2021	Acc	Waist	SMV
[67]	2020	Acc, Gyr	Chest	Max., Min. Peak, Mean, SMV, SD, RMS, etc.
[68]	2020	Acc	Waist	Mean, Kurtosis, Variance, Range, Correlation, etc.
[69]	2020	Acc	Wrist	mean, median, SD, mean absolute deviation, etc.
[51]	2020	Acc, Bar	Chest	Falling interrupt, Hit interrupt, SVMa, etc.
[70]	2019	Acc and Gyr	Waist	Mean, Variance, Kurtosis, SVMa, SMA, etc.
[71]	2019	Acc, Gyr	Chest	Angular velocity
[72]	2019	Acc	Waist	Mean, SD, Range
[73]	2018	Acc, Gyr, Mag	Wrist	Spatial orientation, Movement decomposition
[74]	2018	Acc, Gyr, Mag, Bar	Waist	Distance, Stride time, Coefficient of variation, etc.
[75]	2018	Acc, Gyr	Upper trunk	Signal magnitude, Attitude angle
[76]	2018	Acc, Gyr	Waist	Mean, Variance, Kurtosis, SVM, SMA, etc.
[77]	2018	Acc	Waist	Mean, Kurtosis, SD, Variance, Range, Skewness, etc.
[46]	2018	Acc	Waist	Peak, Variance, and Angles
[78]	2017	FSR, Acc	Foot	Fall index, Pitch, Roll
[79]	2017	Acc	Pocket	Orientation, Velocity, Displacement
[53]	2017	Acc, Gyr	Waist	Total vector acceleration, Angular velocity
[80]	2017	Camera	Waist	HOG, GLBP
[81]	2017	sEMG	Limb, ankle	Absolute Value, Variance, Zero Crossing, etc.
[82]	2017	Acc	Neck, waist, chest	Acceleration magnitude
[83]	2017	Acc, Bar	Neck	SVMa, SMA, etc.
[50]	2017	Acc, Bar	Neck	MAV, MAV Difference
[84]	2016	Acc	Pocket	Resultant acceleration
[47]	2016	Acc	Foot, ankle, arm, waist	Signal magnitude
[52]	2016	Acc, Gyr	Chest	Sum acceleration vector, Angular velocity
[85]	2016	Acc, Gyr	Upper trunk	Acceleration, Trunk angle, Angular velocity
[86]	2016	Acc, Gyr, Mag, Bar	Waist	Yaw, Pitch and roll angles
[87]	2016	Acc	Chest	Acceleration, Inactive sub-periods
[88]	2016	Acc, ToF RTx	N/A	Mean, Mean of Absolute, Variance, Skewness, etc.
[89]	2015	ECG	Not specified	Heart rate variability
[90]	2015	Acc, Gyr, Mag	Waist	Pitch and roll angles, SVMa
[91]	2015	Acc, Gyr	Chest, Waist	Angular velocity, acceleration
[92]	2015	Acc	Wrist	Energy values
[54]	2015	Acc, Gyr, Bar	RAIS	Velocity, Change of posture and height
[93]	2014	Acc, Gyr	Waist	Mean, Variance, Velocity, etc.
[94]	2014	Acc	Head, waist, ankle	SMV, Tilt angle

Table 4 (continued)

Ref.	Year	Sensors	Sens. Pos.	Features
[48]	2014	Acc	Chest, thigh	Not specified
[49]	2014	Acc	Chest, thigh	SVMa
[95]	2014	Acc, Gyr, Mag	Sternum	Trunk angular velocity and angle
[96]	2013	Camera	Waist	HEO and HES
[97]	2013	Acc	Hip, wrist	Mean, SD, Energy, Entropy, Correlation
[98]	2013	Acc	Upper trunk	Resultant acceleration
[99]	2013	Acc, Gyr	Waist, thigh	Trunk sagittal angle and angular velocity
[100]	2012	Acc	Arm	Not specified
[101]	2012	Acc	Hip, shoulder	Total vector acceleration
[102]	2011	Acc	Ankle, waist, chest	Mean, Variance
[103]	2011	Acc, Gyr, Mag	Waist, shank, thigh	Attitude of the shank, thigh, waist, etc.
[104]	2010	Acc	Waist	Area under SMC, SVMa, Reference velocity, etc.
[105]	2010	Acc	Waist	Means, SD, Correlation coefficients, Linear fitting
[106]	2009	Acc, Gyr.	Waist	Not specified
[107]	2008	Acc, Gyr	Waist	Magnitude of inertial frame Vertical velocity
[108]	2006	Acc	Waist	Reverse impact, Average acceleration, etc.
[109]	2006	Acc	Abdomen	Acceleration sequences

Legend: Ref.—Reference; Sens. Pos.—Sensor position; Acc—Accelerometer; Gyr—Gyroscope; Mag—Magnetometer; Bar—Barometer; FSR—Force Sensing Resistor; SMV—Signal Magnitude Vector; SD—Standard Deviation; SVMa—Sum Vector Magnitude; SMA—Signal Magnitude Area; SMC—Signal Magnitude Curve; RMS—Root Mean Square; HOG—Histogram of Gradients; GLBP—Gradient Local Binary Pattern; MAV—Maximum Absolute Value; HEO—Histogram of Edge Orientation; HES—Histogram of Edge Strength; sEMG—Surface Electromyography; RAIS—Right anterior iliac spine; ToF RTx—Time of Flight Radio transceiver; Max.—Maximum; Min.—Minimum

Wearable IMU Systems

Inertial Measurement Units (IMUs) combining accelerometers, gyroscopes, and sometimes magnetometers provide comprehensive motion analysis. For instance, Liu et al. [103] introduced a gravity center detection algorithm with IMU (9-axis) data to distinguish falls from normal activities. Similarly, Pierleoni et al. [86, 90] developed wearable systems incorporating Attitude and Heading Reference Systems (AHRS) and barometers for improved motion and orientation tracking.

Novel Wearable Devices

Innovative FDS designs have emerged, such as footwear-based systems [78], which embed accelerometers and force sensors in insoles to analyze pitch, roll, and fall indices. Additionally, 9-axis IMU-based systems [74, 95] focus on angular velocity and trunk motion to detect falls and assess risk, demonstrating the versatility of wearable FDS.

Camera-Assisted Systems

Camera-integrated wearable systems add a layer of visual analysis. Ozcan et al. [96] utilized edge orientation and optical flow descriptors to detect falls, while their later work [80] incorporated HOG and GLBP features for indoor and outdoor fall detection.

Machine Learning-Based Approaches

Machine learning (ML) methods have significantly transformed wearable Fall Detection Systems (FDS), leveraging data from various sensors to improve detection accuracy, adaptability, and efficiency. This section synthesizes ML-based FDS approaches by focusing on sensor types, data processing methods, feature extraction, and sensor placement while categorizing similar studies.

Accelerometer and Gyroscope Systems

Accelerometers and gyroscopes are fundamental in ML-based FDS for capturing motion dynamics. Aziz et al. [93] utilized a waist-mounted inertial sensor combining a tri-axial accelerometer and gyroscope to analyze pre-impact falls. By examining window sizes and lead times, their study demonstrated how SVM classifiers with an RBF kernel could achieve accurate fall detection. He et al. [75, 85] employed Kalman filters for estimation and KNN classifiers to classify falls, optimizing features such as signal magnitude and altitude angle. Similarly, Shibuya et al. [91] integrated multi-axis accelerometers and gyroscopes in a Wireless Gait Analysis Sensor (WGAS), extracting six ranges of angular velocity and acceleration features for SVM classification. These systems, with sensors placed on the upper trunk or waist, emphasize the importance of precise feature engineering in robust fall detection. Waheed et al. [63] introduced a noise-tolerant fall detection system

Table 5 Publicly available datasets on wearable fall detection systems

Dataset	Simulated/ Real	Indoor/ Outdoor	Age Range (Year)	Health	No. of Fall Type
FallAIIID [110]	Simulated	Outdoor	21 to 53	Healthy	35
KFall [111]	Simulated	Indoor	21 to 28	Healthy	15
tFall [112]	Simulated	Indoor	20 to 42	Healthy	8
MobiAct [113]	Simulated	Indoor	20 to 47	Healthy	4
MobiFall [114]	Simulated	Indoor	22 to 47	Healthy	4
TST FB4FD [115]	Simulated	Indoor	21 to 55	Healthy	32
IDONT-FALL [74]	Simulated	Indoor	N/A	Both	N/A
SisFall [116]	Simulated	Indoor	Adults 19-30, Elderly 60-75	Healthy	15 Fall/ 19 ADL
Smart-phone dataset [117]	Simulated	Indoor	22-42	Healthy	8 Fall
HRIMU [69]	Simulated	Indoor	mean=25.8, SD=3.6	Healthy	6 type falls, 4 near-falls, 9 ADL
WS dataset [118]	Simulated	Indoor	Male (24 ± 3), Female (21.5 ± 2.5)	Healthy	20 falls/ 16 ADL
FARSEE-ING [119]	Real	N/A	N/A	Healthy	N/A
Sisfall 2 [46]	Simulated	Both	60-81	Healthy	N/A

Legend: ADL– Activities of Daily Livings; WS– Wearable Sensor

based on wearable sensors and deep learning, utilizing a bidirectional LSTM model with RNN stacking for processing time-series accelerometer and gyroscope data. Mekruksavanich and Jitpattanakul introduced [60] FallNeXt, a sensor-based fall detection system incorporating accelerometer and gyroscope data with a deep residual model using multi-branch aggregation. Yhdego et al. [62] developed a wearable sensor based real-time fall detection system. The proposed system consists of multi-window segmentation and deep learning models like CNN, BiLSTM, and transformer encoder architectures for accurate classification of falls and near-falls. Ruiz-Garcia et al. [61] developed Care-Fall, a fall detection system using smartwatch-based accelerometer and gyroscope data. The system employs both

threshold-based methods and a range of machine learning algorithms to detect falls in real time. To prioritize critical features in wearable sensor data for fall detection, Sahni et al. [56] proposed an ensemble deep learning framework combining CNN with channel attention and BiLSTM with temporal attention.

Feature-Rich Accelerometer Systems

Sophisticated feature extraction pipelines improve FDS robustness. Nahian et al. [66] proposed a pipeline that extracted over 7,700 cross-disciplinary features from accelerometer data. Feature selection using mutual information and the Boruta algorithm ensured high accuracy (99%) across datasets like UR Fall, MOBIFALL, and UP Fall, employing classifiers such as SVM, Random Forest, and Naïve Bayes. Luo et al. [94] utilized accelerometers placed on the head, ankles, and waist, applying decision tree classifiers to analyze mobility statistics and optimize sensor positions. These studies underscore the value of dominant feature identification and optimal placement for achieving generalized and efficient fall detection. De Raeve et al. [59] designed low-cost Bluetooth Low Energy (BLE) fall detection wearable device, especially for elderly residents in nursing homes. This proposed device uses a threshold-based acceleration and orientation detection while comparing results with a CNN-based classifier. A et al. [57] developed an AI-based wearable system using accelerometer and edge computing boards and deep learning algorithms to detect and prevent falls in elderly people.

Multi-Sensor Fusion Systems

The fusion of data from multiple sensors enriches detection precision. Quadros et al. [73] combined accelerometers, gyroscopes, and magnetometers in wrist-worn devices, contrasting ML and threshold-based methods for evaluating signal effectiveness. Wang et al. [83] integrated barometric sensors with accelerometers, employing wrapper-based feature selection and binary decision tree classifiers to balance fall detection accuracy and energy consumption. Nho et al. [69] incorporated heart rate sensors with accelerometers, leveraging Gaussian Mixture Models (GMM) and a two-stage feature selection process to enhance user adaptability. These studies highlight the benefits of sensor fusion in improving performance.

Advanced ML Architectures

Emerging ML architectures address complex fall detection challenges. Al-qaness et al. [64] introduced MKLS-Net, which integrates multi-kernel convolution, LSTM, and

self-attention mechanisms to analyze spatial and temporal sensor data. This model achieved more than 99% accuracy across MobiAct, SisFall, and UniMib-SHAR datasets, effectively distinguishing falls from similar activities. Yu et al. [65] employed a TinyCNN in a low-power microcontroller, demonstrating 98% sensitivity and specificity on KFall and SisFall datasets by capturing both temporal and spatial motion features. Medrano et al. [79] proposed a novelty detection approach using SVM-based anomaly detectors to classify fall events as deviations from normal patterns, addressing the challenge of diverse fall scenarios. These systems exemplify advancements in real-time, resource-efficient FDS implementations. Guo et al. [58] proposed PIFall, a wearable fall detection system using high-resolution pressure insoles with a 90-sensor matrix and time-series "stress video" analysis, employing a ResNet(2+1)D model for robust classification of elderly activities.

Innovative Sensor Integration

Integrating novel sensors expands the functionality of FDS. Shastry et al. [88] combined IMUs with Time of Flight (ToF) transceivers to enhance fall detection precision by integrating location tracking and reducing false alarms with SVM classifiers. Saleh et al. [72] employed low-complexity accelerometer-based algorithms, utilizing online feature extraction tools to minimize computational burden, and applied SVM for accurate fall classification.

Sensor Placement and Optimization

Optimal sensor placement significantly influences FDS performance. Aziz and Robinovitch [102] examined accelerometer positions across multiple body locations, including the ankles, waist, sternum, and head, analyzing whole-body kinematics using LDA. Yu et al. [82] addressed sensor displacement using an orientation-adjusted HMM algorithm, validated on the FARSEEING dataset, ensuring consistent accuracy across varied conditions. Xi et al. [81] conducted

a comprehensive evaluation of surface electromyography (sEMG) sensor-based fall detection systems by analyzing data collected from sensors placed on the lower limbs of participants during various activities. They assessed the effectiveness of fifteen feature extraction methods and five classification models, including support vector machines (SVM) and k-nearest neighbors (KNN), in identifying fall scenarios. Their findings emphasized the critical role of feature selection and optimal sensor placement in enhancing detection accuracy and system reliability.

Vision Sensor Based Fall Detection Approaches

With the advancement of computational power and computer vision algorithms, researchers utilise vision sensors to detect falls. Similar to the wearable FDSs, the vision-based FDSs also used ML and threshold algorithms. A general structure of the vision-based FDS is shown in Fig. 4. Table 6 gives a comparative summary of vision sensor-based FDSs studied in this work. The publicly accessible vision-based fall datasets were presented in Table 7.

Threshold-Based Approaches

Threshold-based approaches have been extensively utilized in vision-based Fall Detection Systems (FDS), leveraging background subtraction and feature analysis to detect falls. These systems often rely on setting optimal threshold values for extracted features to distinguish fall events from normal activities.

Background Subtraction and Feature Extraction

Rajabi et al. [158] proposed a vision-based FDS using a Gaussian Mixture Model (GMM) for background subtraction to extract moving objects from video sequences. Features such as pixel distance, center of mass width variation,

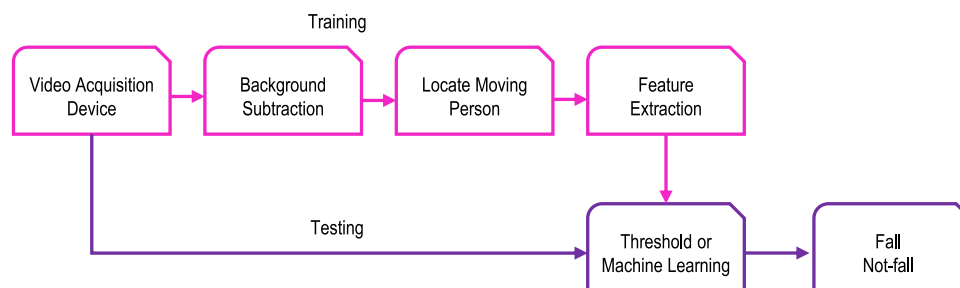


Fig. 4 Vision sensor-based fall detection block diagram where different vision sensors can be used such as surveillance camera, Kinect, depth camera, IR camera. Background subtraction is performed to

remove static background. After background subtraction, the image has passed through some preprocessing steps, feature extraction steps, and classification steps to detect falls

Table 6 Sensors and preprocessing technique for vision-based fall detection approaches

Ref	Year	Sensor	POI to be tracked	Preprocess
[120]	2025	RGB Cameras	Human	Resolution Filter, Resizing (416×416), Normalization, Focus on CBH
[121]	2025	RGB Camera	Skeleton joints	YOLOv8, BoT-SORT, Sliding Window, Post-Fall Removal, Min–Max Norm
[122]	2025	RGB Camera	Skeleton joints	YOLOv8, AlphaPose, Angle & Velocity Calc
[123]	2025	RGB Cameras	Moving object; Human posture and motion	BS, Foreground Extraction, DataAug, Normalization, Noise Reduction
[124]	2024	RGB Camera	Skeletal keypoints	Multi-Res 3D Conv, Spatio-Temporal Pooling, Pseudo-3D
[125]	2024	RGB, IR, Depth, Thermal	Whole-frame anomaly	Grayscale, Resizing (64×64), Normalization, Frame Rate reduction
[126]	2024	Raspberry camera; Jetson Nano	Human body	Manual Labeling (Roboflow), Quantization, Pruning
[127]	2024	RGB Cameras	Human activities	Sliding Window, Resizing, Normalization
[128]	2024	RGB Webcam	Skeleton keypoints	BS, Otsu, Morph Filter, BlazePose
[129]	2023	RGB Camera	Human body	Grayscale, CNN Segmentation, 3-level Fusion
[130]	2023	RGB Camera	Skeleton joints	OpenPose, Frame Filter, View Norm., Seq Fix, ZMUV Norm.
[131]	2022	RGB Camera	Full body posture and motion	ACB Conv, Spatial Attention, Multi-Scale Pred., Anchor Clustering
[132]	2024	Pi Camera	Human body	Video segmentation, Frame extraction, Background subtraction, Noise removal using correlation filters
[133]	2023	Camera	human body	Optical Flow Features Extraction
[134]	2022	360 Camera, RGB Camera	human body	Frame sampling, Resizing to 224 × 224 pixels, Dynamic image generation using rank-pooling, Normalization of pixel values, Data augmentation (rotation, scaling, flipping)
[135]	2021	Camera	human body	Histogram of Optical Flow, Motion Boundary Histogram, Fall Motion Vector
[136]	2020	Camera	Human Posture	Optical flow Algorithm
[137]	2019	Depth Camera	Human Posture	VGG16 used to obtain feature map
[138]	2019	Kinect	3D skeleton	Segmentation, labelling, Searching
[139]	2019	Camera	Human Body Key Point	Deep Cut NN used to get key point map
[140]	2019	Fisheye camera	Human Posture	Background Subtraction, Synthetic Data generation
[141]	2019	IR Camera	Person	Analysing Different Background Subtraction algorithm
[142]	2019	RGB Camera	Person	Obtain key points of human body
[143]	2018	Kinematic Camera	Person	N/A
[144]	2018	Depth Camera	Head, upper part of Body	Head and Upper part of Body Identification
[145]	2018	Depth Sensor	floor plan and skeleton	N/A
[146]	2018	Camera	Moving Object	N/A
[147]	2018	Camera	Human Posture	BS, AET
[148]	2017	Pi camera	Human motion	Segmentation, AET
[149]	2017	Camera	Moving Object	Optical flow algorithm, Noise Reduction
[150]	2017	Pi camera	Human Posture	GMM for BS
[151]	2017	RGB Camera	Human Posture	Segmentation
[152]	2017	RGB Camera	Person	RCNN employed to detect person
[153]	2016	Single camera	Humane posture	MOG for BS, AET
[154]	2016	Depth Camera	Head, Centre of Mass	Head tracked using AET and particle filter algorithm
[155]	2016	Surveillance Camera	Human Shape	BS
[156]	2016	N/A	Person	SOV. Bag of Word method extract feature from SOV
[157]	2015	Camera	Human Blob	Accumulated computing approach for segmentation
[158]	2015	Camera	Moving Object	BS
[159]	2015	IP camera	Human Posture	Foreground Segmentation
[160]	2015	Kinect	Person	N/A
[161]	2014	Depth Camera	Humane Posture	BS, Segmentation
[162]	2014	Depth Camera	Human Shape	BS
[163]	2014	Camera	Human Posture	Joint routing and homography transformation
[164]	2014	Kinect	Key Joint	RDT used to acquire human key joint
[165]	2014	Video Camera	Person	Median filter used to get silhouette
[166]	2013	Single USB Camera	Human Posture	Codebook BS
[167]	2013	Surveillance Camera	Human Posture	Bag of Word

Table 6 (continued)

Ref	Year	Sensor	POI to be tracked	Preprocess
[168]	2013	Monocular Camera	Human Shape	BS
[169]	2013	Kinect, Overhead Camera	Human Shape	AET
[170]	2012	Two camera	Voxel Person	Codebook BS
[171]	2012	Camera	Moving Person	Silhouette area extracted by Running average Method
[172]	2012	Stereo Vision	Human Shape	N/A
[173]	2012	Webcam	Person	Foreground -Background Subtraction
[174]	2012	Kinect	Head, Shoulder	Head detection from moving object
[175]	2011	Single USB Camera	Human Posture	Codebook BS, AET
[176]	2011	Camera	Human Shape	Background Subtraction, Canny Edge detector
[177]	2011	IP camera	Person	BS
[178]	2010	ToF Camera	Human Posture	Segmentation, BS
[179]	2009	IP camera	Human Shape	BS
[180]	2008	Camera	Human posture	N/A
[181]	2008	Surveillance Camera	Human Posture, Head	BS, AET
[182]	2006	Camera	Person	BS

Legend: POI–Point of Interest; NN–Neural Network; BS–Background Subtraction; AET–Approximated Ellipse Fitting Technique; GMM–Gaussian Mixture Method; MOG–Mixture of Gaussian; RDT–Randomise Decision Tree; SOV–Silhouette of Variation; RCNN–Regional Convolutional Neural Network; ToF–Time of Flight

Table 7 Publicly available datasets on vision-based fall detection systems

Ref	Dataset	Year	Frame Rate	Sens. Pos.	Simulated/ Real	Environment
[183]	URFD	2014	30 fps	Ceiling	Simulated	Indoor
[162]	SDUFall DATASET	2014	30 fps	Wall	Simulated	Indoor
[184]	NTU RGB+D	2016	15 fps	N/A	Simulated	Indoor
[185]	Mulicam Fall Dataset	2010	30 fps	Ceiling	Simulated	Indoor
[186]	Li2e	2013	25 fps	Wall	Simulated	Indoor
[187]	Sports-1M,	2014	30 fps	N/A	Simulated	Both
[188]	PIROPO	2016	10 fps	Ceiling	Simulated	Indoor
[189]	WEIZMANN ACTION DATASET	2005	N/A	N/A	Simulated	Indoor
[190]	UCF11	2009	30 fps	N/A	Both	Outdoor
[191]	HMDB-51	2011	N/A	N/A	Movie, YouTube	Both

Legend: Sens. Pos.– Sensor Position

Hu moment invariants, and angle were analyzed, and an optimal threshold was applied to identify falls. A finite state machine (FSM) was incorporated to minimize detection time, effectively addressing occlusion and overlapping issues.

Similarly, Nguyen et al. [153] employed motion history and changes in human shape features extracted via Mixture of Gaussian (MOG) and Ellipse Fitting techniques. Thresholds were applied to these features to monitor falls from pre-fall to post-fall phases. Albawendi et al. [147] extended this idea by integrating Time Motion History Image (tMHI) analysis and change in human shape orientation, utilizing three wall-mounted cameras for feature extraction. Thresholds on motion, shape orientation, and projection histograms enabled efficient fall detection. Wang and Deng [128] presented a lightweight and real-time fall detection algorithm combining background subtraction and BlazePose pose estimation with a cascaded random forest classifier for enhanced efficiency on low-power systems.

3D and Multi-Camera Systems

Leone et al. [178] introduced a 3D fall detection system using a Time-of-Flight (ToF) camera, where Bayesian segmentation and Kalman filters tracked human movement. Thresholds on 3D centroid distances effectively detected falls, with additional analysis of four distinct human poses. Auvinet et al. [177] tackled occlusion challenges using multiple cameras to compute the Vertical Volume Distribution Ratio (VVDR), a feature differentiating lying on the floor from other activities. While effective, these methods struggled with scenarios involving furniture interactions.

Foreground-Background Segmentation

Foreground segmentation techniques have also been pivotal. Joshi and Nalbalwar [150] developed a system using Gaussian Mixture Models to extract movement data. Features like aspect ratio, centroid, orientation, and Hu moments were

thresholded to detect falls in real-time. Similarly, Nghiem et al. [174] focused on head detection through modified Histogram of Oriented Gradient (HOG) features to locate heads among moving objects, enabling accurate fall identification. Edgcomb et al. [173] addressed privacy concerns by using bounding ovals and Minimum Bounding Rectangles (MBR) for feature extraction, employing Dynamic Time Wrapping (DTW) to compute thresholds for fall detection. This approach highlighted the importance of temporal feature analysis in privacy-sensitive environments.

Low-Cost and Depth-Based Systems

Wahid et al. [148] proposed a cost-effective FDS using a Pi camera and Raspberry Pi module to reduce system complexity. Motion History Images (MHI) and motion speed were analyzed using threshold algorithms, though human intervention was needed to address false alarms. Nizam et al. [145] employed a Kinect sensor to calculate fall risk levels, using skeleton tracking and velocity thresholds to adapt detection to user-specific risk profiles. Depth imaging has also been utilized, as demonstrated by Merrouche and Baha [154], who extracted human shapes from depth frames and analyzed head motion using covariance models. Covariance-based thresholds effectively separated falls from normal activities while minimizing computational overhead.

Other Threshold-based Vision FDS

Kroputaponchai et al. [179] developed a system leveraging Motion Mixture Models to track human shapes across frames, examining changes in human shape features to identify falls. Kottari et al. [140] utilized fisheye cameras to capture segmented silhouettes and applied cosine angle-based thresholds for fall detection, showcasing the effectiveness of low-cost hardware in indoor settings. Fuzzy rule-based systems further expanded the scope of threshold-based approaches. Serrano-Cuerda et al. [157] employed fuzzy logic to process geometrical and kinematic features extracted from segmented human blobs, achieving robust detection of falls and associated patterns.

Machine Learning-Based Approaches

Traditional Machine learning (ML) techniques have been widely adopted in vision-based Fall Detection Systems (FDS), leveraging advanced algorithms and diverse sensors to improve detection accuracy and robustness. This section synthesizes key contributions by grouping studies into subcategories based on methodologies, sensor types, and computational approaches.

Depth Camera-Based Systems

Depth cameras have gained prominence in vision-based FDS for their ability to preserve privacy and operate in varying lighting conditions. Zhao et al. [144] employed a depth camera to monitor falls from beds, using Random Forest (RF) algorithms to extract human head and upper body motion features. The Large Margin Nearest Neighbor (LMNN) algorithm was then applied to classify falls versus Activities of Daily Living (ADLs).

Hybrid Depth and RGB Systems

Hybrid approaches that combine depth and RGB data focus on integrating complementary information from both modalities to enhance fall detection. Josemans et al. [169] demonstrated effective fusion of skeleton-based depth features with RGB motion features, showing improved detection accuracy through Principal Component Analysis (PCA) and SVM classification. Bian et al. [164] employed an infrared depth camera to estimate human key joints using a Randomized Decision Tree (RDT) algorithm and used these features in conjunction with trajectory analysis for fall classification via SVM. These approaches leverage the strengths of both depth and RGB data to address challenges in dynamic environments.

Single RGB Camera Systems

Single RGB camera systems offer simplicity and cost-effectiveness. Yu et al. [175] developed a semi-supervised fall detection model using a digital camera positioned near the ceiling to track individuals. Background subtraction via the codebook method was employed to extract human silhouettes, from which features were computed and applied to an Online One-Class SVM (OCSVM) for real-time fall detection. Two thresholds were introduced to reduce false negatives, with the model continuously updated to personalize detection for individual users. Building on this, Yu et al. [175] applied a similar setup for human posture recognition using a Directed Acyclic Graph SVM (DAGSVM). Global and local features, such as histograms and ellipse-fitting parameters, were extracted and analyzed to classify human postures, enabling fall detection through predefined conditions. Kaur et al. [132] proposed a hybrid system combining Haar Cascade for human detection and background subtraction for motion analysis. Their system demonstrated high accuracy on UMAFall and LE2I datasets, addressing robustness and efficiency challenges in fall detection systems. Mirmahboub et al. [171] utilized silhouette area changes detected via the Running Average Method and integrated these features with an SVM classifier for fall

detection. Makantasis and Doulamis [168] extended this approach by incorporating 3D space approximations from monocular camera data, using SVM with a Gaussian Radial Basis kernel to classify falls. Harrou et al. [151] combined Generalized Likelihood Ratio (GLR) with SVM to distinguish falls from fall-like activities, leveraging body segmentation into predefined regions for feature extraction.

Multi-Camera Systems

Multi-camera setups enhance system coverage and robustness. Wang et al. [167] proposed a multi-camera FDS using surveillance footage, extracting MOSIFT features and employing Bag-of-Words representations alongside SVM classifiers for fall detection. Hanson et al. [163] introduced a decentralized approach with intelligent camera networks, combining inter-image homographies and Cyclops algorithms for fall detection and localization.

Advanced Feature Extraction and Classification

Innovative algorithms have been developed to address computational and accuracy challenges. Ma et al. [162] utilized Kinect depth cameras to extract Curvature Scale Space (CSS) features and applied the Bag-of-Words method with an Extreme Learning Machine (ELM) for classification. Similarly, Akagündüz et al. [156] introduced the Silhouette Orientation Volume (SOV) descriptor, employing K-medoids clustering for feature extraction and Naive Bayes for classification. These approaches demonstrated efficacy across diverse datasets, including SDU-Fall and Weizmann Action.

Skeletal Feature-Based Systems

Recent works have explored skeletal feature-based detection. Alaoui et al. [142] utilized 2D skeleton models to analyze key point positions, applying classifiers such as SVM, Decision Trees (DT), and Random Forests (RF) to distinguish fall incidents. Alazrai et al. [160] extended this concept with Kinect sensors to extract Motion Pose Geometric Descriptor (MPGD) features from skeletal joints, training SVM classifiers for fall event detection.

Silhouette and Privacy-Preserving Methods

Feature extraction remains central to ML-based FDS. Rougier et al. [176] leveraged shape deformation analysis using Gaussian Mixture Models (GMM) to identify falls through silhouette shape changes. Anderson et al. [182] employed Hidden Markov Models (HMM) with silhouette-based features for privacy-preserving fall detection. Alonso et

al. [141] compared 30 background subtraction algorithms, identifying LBAdaptiveSOM as the most effective for nighttime fall detection, emphasizing the importance of algorithm optimization.

These studies collectively highlight the advancements in ML-based vision FDS, demonstrating their ability to address diverse challenges such as privacy, computational complexity, and environmental variability. Future research should focus on refining hybrid systems, enhancing temporal feature analysis, and integrating advanced deep learning techniques for improved robustness and scalability.

Shallow Neural Network-based Approaches

Neural network-based systems leverage multi-layer perceptrons (MLPs) and bio-inspired architectures for fall detection. Humenberger et al. [172] proposed a bio-inspired FDS using stereo vision sensors to generate 3D representations of monitored spaces. Extracted features were classified using multiple neural networks, demonstrating robustness through a voting mechanism. Similarly, Alhimale et al. [165] utilized silhouettes normalized from video frames as inputs to a neural network, combining FSM models and thresholds for effective fall detection. Shen et al. [139] utilized a cloud-based neural network system with Raspberry Pi cameras, leveraging body position data for fall classification. Núñez-Marcos and Arganda-Carreras [127] proposed the first pure RGB-based transformer approach for fall detection to process 16-frame clips without the need for optical flow or skeleton extraction. The proposed model leverages the Uniformer backbone, comprising Dynamic Position Embedding, Multi-Head Relation Aggregator, and feedforward layers.

Foroughi et al. [180] utilised the Integrated Time Motion Image (ITMI) and Eigenspace method to obtain an efficient elderly FDS which used a neural network for fall detection purposes. The temporal motion properties of video sequences were extracted using ITMI and then projected to Eigenspace to reduce the feature vector. A four-layer MLP neural network was applied to detect falls and other everyday activities using extracted features as input.

Like [180], the temporal motion properties were further utilised by Foroughi et al. [181] for developing a vision-based fall detection system. Surveillance cameras were implemented to capture video data. Background segmentation was undertaken by quantifying the current frame's difference with a reference frame to obtain a human silhouette. The approximated ellipse fitting method was applied to the extracted silhouette to retrieve human shape features. The head movement information was extracted from a

rectangular bounding box for each frame. An MLP neural network was implemented to classify different types of normal, abnormal, and fall activities. Temporal motion properties are central in systems like those of Foroughi et al. [180], where Integrated Time Motion Image (ITMI) and Eigenspace projections reduced feature dimensionality for classification by an MLP neural network. Extending this, Foroughi et al. [181] incorporated head movement features extracted from bounding boxes, achieving refined classifications for normal, abnormal, and fall activities.

Deep Learning-Based Approaches

Deep learning techniques have significantly enhanced vision-based Fall Detection Systems (FDS), enabling the extraction and analysis of complex spatio-temporal features.

Convolutional Neural Network (CNN)-Based Systems

CNN-based systems dominate recent advancements due to their ability to process high-dimensional visual data. Min et al. [143] extended CNN applicability by integrating Faster R-CNN for spatial analysis, specifically focusing on interactions between individuals and furniture.

Zhang et al. [137] tackled complex environments using trajectory-weighted deep convolutional rank pooling descriptors, combining CNN feature maps with trajectory attention to classify fall events effectively. Tsai and Hsu [138] focused on real-time efficiency by pruning CNN architectures for 3D skeleton data processing. Vishnu et al. [135] integrated motion descriptors with 3D-CNNs to model spatio-temporal dynamics, achieving superior performance across multiple datasets.

Transfer learning approaches, such as those by Nunez-Marcos et al. [136], used pre-trained CNN layers (e.g., VGG-16) to extract optical flow features for detecting falls in publicly available datasets, demonstrating the efficacy of leveraging existing architectures. Alanazi et al. [129] proposed a vision-based fall detection system that uses human segmentation, 3-level image fusion, and a four-stream 3D CNN architecture to accurately detect different stages of human falls in video sequences. Zhang et al. [124] developed a vision-based fall detection model using a high-resolution network (HRST) and a spatio-temporal coordinate attention mechanism to effectively differentiate fall-like behaviors from actual falls through multi-scale 3D convolution and skeletal tracking. Denkovski et al. [125] introduced a novel Temporal Shift loss for anomaly-based fall detection using 3D convolutional autoencoders and multi-modal video input. The proposed system performs the fall prediction and reconstructs overlapping video frames for improved temporal modeling.

Hybrid Deep Learning Systems

Hybrid systems combine neural network architectures with traditional methods or integrate multiple neural approaches for enhanced detection accuracy. Lu et al. [146] proposed a 3D CNN and LSTM combination, extracting spatial and temporal features with attention models to refine frame-specific regions of interest. Saurav et al. [134] designed a dual-stream fused neural network combining CNNs and SAB-BiLSTMs for RGB and 360-degree videos. Predictions from Dynamic-Fall-CNN and SAB-BiLSTM were fused, achieving state-of-the-art accuracy on diverse datasets like Fall360 and UP-Fall.

Hybrid traditional-deep systems include Feng et al. [152], who combined Region-based CNNs for object detection with motion history images (MHI) and histogram of oriented gradients (HOG) features for fall classification using SVMs. Debard et al. [159] integrated background subtraction with particle filters, improving segmentation robustness by tracking foreground objects over time. Zahan et al. [130] proposed a privacy-preserving vision-based fall detection system that extracts 2-D skeletons via OpenPose and fuses joint and motion information into a unified high-dimensional representation. This data is processed through spatial graph convolutions and separable temporal convolutions with randomized masking to capture discriminative spatio-temporal patterns.

YOLO-Based Systems

Chen et al. [131] proposed a fall detection model based on improved YOLOv5s with asymmetric convolution blocks (ACB) and a spatial attention mechanism to enhance human feature localization and fall behavior recognition accuracy. Elwaly et al. [126] designed a novel eldercare robot integrating indoor path planning using Hector SLAM and fall detection using a YOLOv7-based object detection model, deployed on a Jetson Nano with real-time processing capabilities. Sanjalawe et al. [123] proposed a hybrid deep learning fall detection system using YOLOv8 and Time-Space Transformers for real-time spatial detection, modeling temporal motion. The proposed model is evaluated on two publicly available datasets (CUCAFall and Diverse-FALL10500). Liu et al. [122] proposed HFDmia-Pose, a hybrid vision-based fall detection framework combining a modified YOLOv8s detector with AlphaPose and a rule-based algorithm. The proposed model evaluates skeletal velocities and posture angles to detect falls in real time. Yu et al. [121] proposed a YOLOv8 based real-time fall detection system, focusing on privacy preservation. The system uses YOLOv8 for skeleton extraction and BoT-SORT for tracking and train a hybrid Temporal Convolutional Network

and Transformer Encoder (TCNTE) with normalized joint sequences. Zhang et al. [120] introduced YOLO-fall, a lightweight fall detection model. This model combines a pruned ESNet-L backbone with H-Swish activations, re-parameterized RepConv layers, and a C2-based detection head featuring 5×5 depth-separable convolutions, for optimizing both accuracy and computational efficiency.

Ambient Sensor-Based Fall Detection Approaches

Apart from wearable and vision-based fall detection systems, researchers utilise different types of ambient sensors or context-aware sensors to develop efficient FDSs. Several ML algorithms and rule-based methods have been employed over the last few years to detect falls from ambient sensor-based systems. A general structure of the ambient sensor-based FDS is shown in Fig. 5. Table 8 gives a comparative summary of ambient sensor-based FDSs studied in this work. Moreover, the used public datasets on ambient sensor-based FDS are described in Table 9.

Threshold-Based Approaches

Threshold-based approaches in ambient Fall Detection Systems (FDS) leverage predefined criteria to identify falls by analyzing signal patterns and physical parameters.

RFID and Radio-Based Systems

RFID and radio signal-based approaches utilize signal properties to detect falls through spatial tracking and signal analysis. Paolini et al. [208] proposed an RFID-based fall detection system combining a 2D antenna array and Monopulse RADAR for object localization. By measuring azimuthal and elevation angles and applying thresholds to the RFID tag height, the system achieved robust indoor

tracking and fall detection. Borhani et al. [211] introduced a non-wearable radio-based system using a 3D stochastic trajectory model to analyze mobility patterns via Doppler features. Their work emphasized modeling signal characteristics to enhance the reliability of fall detection algorithms, demonstrating that signal envelopes alone are insufficient for fall phase analysis. Doppler power spectral density (PSD) and spectrogram analyses were utilized to identify fall patterns effectively. Newaz and Hanada [192] developed a privacy-preserving fall detection system using a low-resolution infrared array sensor on a Raspberry Pi-based setup. The proposed system detects falls by comparing thermal signatures of human activities with statistical distributions using Earth Mover's Distance.

Acoustic Sensor-Based Systems

Li et al. [215] enhanced the performance of an earlier acoustic fall detector (FADE) [219] by addressing challenges in interference-heavy environments. Using Blind Source Separation (BSS) and Non-negative Matrix Factorization (NMF), the system separated interference from fall signals and reduced background noise. The combination of single-channel BSS with joint NMF improved detection accuracy, especially in complex acoustic scenarios.

Machine Learning-based Approaches

Traditional Machine learning (ML) and deep learning (DL) approaches, often combined with advanced signal processing techniques, have significantly enhanced the robustness and accuracy of ambient Fall Detection Systems (FDS).

Wireless Signal-Based Systems

Wireless signal-based methods utilize channel properties to detect falls through non-invasive monitoring. These systems, such as WiFall [213], exploit Channel State

Fig. 5 Ambient sensor-based fall detection block diagram where different ambient and environmental sensors can be used such as Infrared sensor, vibration sensor, floor sensor. Sensor-recorded data can be passed through some pre-processing, feature extraction and classification steps

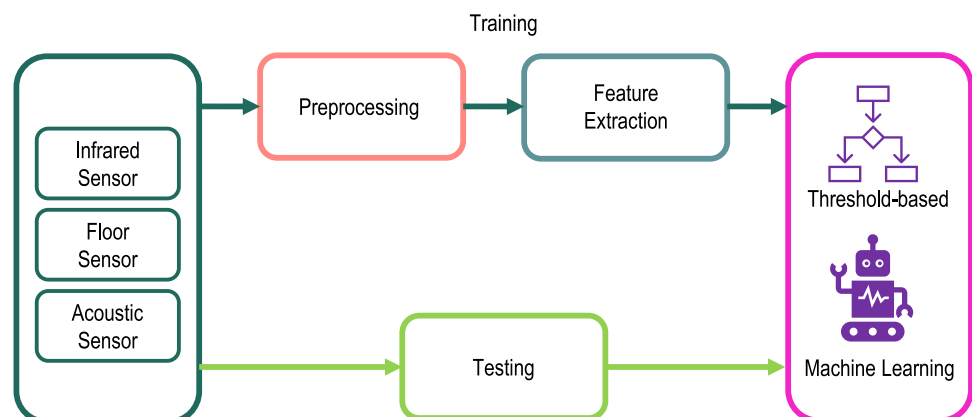


Table 8 Sensors and features used in ambient sensor-based fall detection approaches

Ref.	Year	Sensor	POI to be tracked	Features
[192]	2025	IR Sensor	Human activities	Privacy-preserving, Cost-effective, Real-time, EMD, Thermal Pixel Sum, Stat. Histogram
[193]	2024	N/A	Human motion	PBC, Eff. Acc (Max, Mean, Var)
[194]	2023	Directional antenna	Distortions in LoRa signal reflections	ERF, SSE, KLC
[195]	2023	RFID	Human motion	Cal. Phase, Smoothed RSSI, Var. Seg., Norm. Multi-signals
[196]	2023	Geophone sensors	Floor vibration	Dual-Threshold, DWT (8-level, db4), 100-dim Vec, EoA Ratio
[197]	2023	FMCW mmWave radar	Human	1D PC (z-axis), Doppler Velocity
[198]	2023	PIR	Human motion	Signal Intensity (ADC), Speed, Displacement
[199]	2023	WiFi CSI (Channel State Information)	Human motion	STFT-CSI, Motion Intensity, Recon. Error
[200]	2022	IR Sensor	Human presence and fall events	Centroid, Spee, Area, Frame Var.
[201]	2022	Acoustic sensor on Misty II robot	Ambient sound	N/A
[202]	2021	Acoustic sensor	Unattended household sounds	Spectrogram, Polygram, Tempogram, LBP Texture
[203]	2021	Speaker, Acoustic sensor	Reflected Doppler shift signals	Spectrogram, Doppler Shift
[204]	2024	Audio players, Acoustic sensor	Ambient sound	MFCCs, Speed Characteristics, Extreme Ratio Curve, Spectral Entropy
[205]	2023	Acoustic sensor	Sound events related to subjects' actions	Spectrograms, MFCCs
[206]	2022	Acoustic sensor	Audio signals from bathroom	Log mel spectrograms, Diff features
[207]	2021	Acoustic sensor, Speaker	Human motion	Movement speeds, Spectral entropy, Extreme frequency ratio curves
[208]	2019	RFID, Antenna	Person	Maximum Power Ratio, RSSI.
[209]	2019	Thermal Vision Sensor	Person	Heat map of the area ,change of blobs
[210]	2018	Depthh Sensor	Human	Minimum height of the given joint and Max joint vertical velocity
[211]	2018	Radio Signal	Human Body	Doppler PSD ,spectrogram of channel
[212]	2014	RADAR	Torso, Leg, Arm	STFT, Spectrogram
[213]	2014	Wireless Trasciever	N/A	CSI Amplitude
[214]	2014	PIR, Flex sensor	Person	N/A
[215]	2013	Acoustic Sensor	Acoustics Signal	N/A
[216]	2012	MW,PIR,Pressure Matt	Person	Doppler shift measurements,
[217]	2010	Floor sensor	Person	Number of observation, Longest Dimension, Magnitude etc.
[218]	2009	Acc, Microphone	N/A	Shock response spectrum and MFCC spectrum
[219]	2008	Acoustic Sensor	N/A	MFCC features

Legend: POI–Point of Interest; Acc–Accelerometer; MW–Microwave; PIR–Passive Infra-red; STFT–Short Term Fourier Transform; CSI–Channel State Information; RSSI–Received Signal Strength Indicator; PSD–Power Spectrum Density; MFCC–Mel Frequency Cepstral Coefficient

Table 9 Publicly available datasets on ambient sensor-based fall detection approaches

Ref.	Dataset	Year	Sensor	Sens. Pos.	Age Range	Health	Environment	No. of Fall/ ADL
[220]	TigerPlace	2012	Kinect, Acoustic	Above the door	31-39 Y	Healthy	Indoor	120/120
[221]	fall detection dataset	2019	Infrared	Wall at a height of 1.2 m	27-37	Healthy	Indoor	15
[222]	CAD-60	2011	Microsoft Kinect	1.62–1.97m Height	22-39 Y	Healthy	Indoor	60 ADL

Legend: Sens. Pos.– Sensor Position; Y–Year; ADL– Activities of Daily Living

Information (CSI) to differentiate between fall and non-fall events. By employing sensing, learning, and alerting stages, WiFall dynamically updates its classification model using real-time feedback, reducing false alarms and adapting to changing conditions. Similarly, Borhani et al. [211] advanced radio-based detection by modeling mobility patterns through 3D stochastic trajectories, highlighting the importance of Doppler features for capturing the dynamics of fall phases. Together, these systems underscore the role of wireless signal processing in enabling non-intrusive fall detection. Yang et al. [200] proposed a privacy-preserving, real-time fall detection system that mounts an MLX90640 infrared array sensor on a side wall to capture temperature matrices, applies overlapping sliding windows to extract four key features (centroid shift, velocity, area variation, and temporal variance), and employs an RBF-kernel SVM to distinguish falls from daily activities. Zhang et al. [194] presented LoFall, a fall detection system using LoRa. The system applies background subtraction and bandpass filtering to obtain clear spectrograms, narrows candidate fall segments via Human Movement Curve (HMC) and Energy Mutation Curve (EMC), extracts ERF, SSE, and KLC features, and fuses them with fuzzy-entropy weights before classification by a Random Forest model. Zhao et al. [195] proposed a wear-free, device-independent fall detection framework that leverages a passive RFID tag array to capture RSSI and phase signals. The model then applies phase calibration, wavelet-based denoising, and variance-driven segmentation to extract action snippets, and classifies ten activities using a deep residual network adapted for 1D time-series data. Wu et al. [196] presented G-Fall, a device-free, user-independent fall detection system that senses floor vibrations via geophone arrays, segments events with an energy-based dual threshold and extracts multiresolution DWT features. Based on the collected data, the proposed model can classify falls using a Hidden Markov Model, with Energy-of-Arrival reconfirmation to suppress false alarms. Kittiyapunya et al. [197] proposed a privacy-preserving fall detection system using 1D point cloud data and Doppler velocity derived from a mmWave FMCW radar, employing an LSTM network for robust classification in indoor environments.

Hassan et al. [198] presented a low-cost, non-intrusive fall detection framework using wall-mounted PIR sensors integrated with Arduino-based hardware and classical machine learning classifiers to detect falls in elderly individuals with high accuracy. Ji et al. [199] proposed a practical real-time fall detection system “SiFall”, that uses WiFi-based RF sensing and a self-supervised variational autoencoder (FallNet). SiFall system is trained on STFT-transformed CSI data to detect falls as anomalies. A low-cost real-time fall detection system was proposed by Arnaoutoglou et al. [193], that used

CW radar for fall detection. The proposed system extracts Doppler-based effective acceleration features and applies lightweight SVM and neural network classifiers for classification of fall events.

Acoustic and Audio-Based Systems

Acoustic approaches leverage audio and vibration signals for unobtrusive fall detection. Amsaprabhaa [205] and Popescu et al. [219] highlighted the utility of Mel Frequency Cepstral Coefficients (MFCC) features for effective classification, while employing noise reduction techniques like Wiener filters to enhance reliability. Zigel et al. [218] expanded on these methods by integrating Shock Response Spectrum (SRS) features with MFCC in a two-stage Bayesian classification framework, demonstrating strong performance in simulated fall scenarios. Chen et al. [204] further advanced acoustic detection with FA-Fall, a multimodal system combining active and passive sensing. By leveraging LSTM-based cross-modal attention, FA-Fall achieved robust performance even in noisy environments, addressing privacy concerns and enhancing adaptability across settings. Collectively, these works emphasize the versatility of acoustic signals for privacy-preserving and robust fall detection. Pandya et al. [202] presented a novel Ambient Acoustic Event Assistive Framework that records unattended household sounds via a network of microphone sensors, transforms audio into 2D spectrogram images, extracts texture features with Local Binary Patterns, and classifies 22 types of events using a hybrid LSTM-CNN model alongside baseline ML classifiers. Lian et al. [203] developed a fall detection system using inaudible acoustic sensing through a home audio device emitting ultrasonic waves and analyzing Doppler shift signals captured by a microphone. The system leverages Hidden Markov Models for classification, optimized with Singular Value Decomposition and K-means clustering. Kaur et al. [201] introduced a Transformer-based fall detection system using ambient audio signals captured in bathroom environments, aiming for a non-intrusive, privacy-preserving, and scalable solution deployed via an autonomous mobile robot.

Specialized Systems for Unique Modalities

Specialized sensing modalities provide innovative pathways for fall detection. Lian et al. [207] utilized inaudible acoustic signals and Doppler shifts to achieve lightweight and privacy-conscious fall detection, employing Hidden Markov Models (HMM) for effective classification. Similarly, floor sensor systems like Rimminen et al. [217] applied Near Field Imaging (NFI) to analyze mobility patterns,

employing Bayesian filtering for pose estimation. These approaches highlight the potential of leveraging unique physical interactions and signal properties for non-intrusive detection.

Threshold-based and machine learning hybrid approaches further exemplify the adaptability of these systems. Wu et al. [212] combined thresholding techniques with ML classifiers to analyze Doppler signatures from radar signals. By extracting features such as frequency magnitudes and event lengths, their sparse Bayesian model effectively classified fall events under diverse conditions. Rafferty et al. [209] extended this hybrid concept with thermal vision systems, integrating threshold-based blob detection and CNN-based data-driven analysis. While effective, their findings emphasized the challenges of small training datasets in achieving robust CNN performance.

Multimodal Fall Detection Approaches

A multimodal FDS involves the fusion of wearable sensors, vision sensors, and ambient sensors. The fusion may take place at many different levels, including feature level, data level, and decision level. Figure 6 represents a general architecture of multimodal FDS. Similar to the above approaches, the multimodal FDS also employs threshold-based and machine-learning algorithms. Table 10 gives a comparative summary of multimodal FDSs studied in this work. Also, Table [11] summarizes the multimodal publicly available fall detection datasets.

Threshold-Based Approaches

Threshold-based approaches in multimodal FDS capitalize on the complementary strengths of different sensors to improve detection reliability and accuracy. Systems leveraging wearable sensors, such as accelerometers, are often combined with ambient or vision-based sensors to provide robust fall detection solutions through data fusion at various levels.

Wearable sensors, such as tri-axial accelerometers, are frequently used for their ability to capture motion data, while heart rate monitors and photoelectric sensors add contextual information. For instance, systems like those proposed by Makhoul et al. [223] integrate these modalities using feature-level fusion to monitor falls, heart rates, and locations. Decision-level fusion then acts as a critical layer to trigger notifications to emergency services, showcasing the effectiveness of hierarchical fusion techniques.

Incorporating vision sensors adds another dimension to fall detection, allowing systems to account for spatial and visual data. Studies such as those by Cippitelli et al. [224] and Kwolek and Kepski [225] demonstrate the synchronization of accelerometer data with Kinect depth images. These systems apply thresholds on parameters such as joint positions and acceleration magnitude, while fuzzy logic further validates fall events, emphasizing the benefits of combining logical reasoning with physical data thresholds.

Systems combining wearable and vision sensors are particularly adept at addressing the limitations of individual modalities. For example, Fernández-Caballero et al. [226] effectively utilized accelerometers for outdoor fall detection and vision sensors for indoor scenarios. This dual approach allowed the system to adapt to diverse environments, merging outputs from both modalities to enhance reliability. Similarly, Ozcan and Velipasalar [227] proposed a system that fused accelerometer data with camera-based features like EO Histograms and GBLP, reducing false positives and ensuring high detection accuracy through complementary validation.

These threshold-based multimodal systems collectively demonstrate the potential of combining wearable, vision, and ambient sensors to create robust fall detection frameworks. The integration of thresholds with hierarchical fusion techniques across multiple modalities ensures improved accuracy and reliability, addressing the limitations inherent in single-sensor systems.

Fig. 6 Multimodal fall detection block diagram. A fusion of vision, wearable and ambient sensors can be made to develop a multimodal fall detection system. Features are extracted from each approach and then merged for feature-level fusion. After merging, ML or threshold-based algorithm is deployed to detect falls

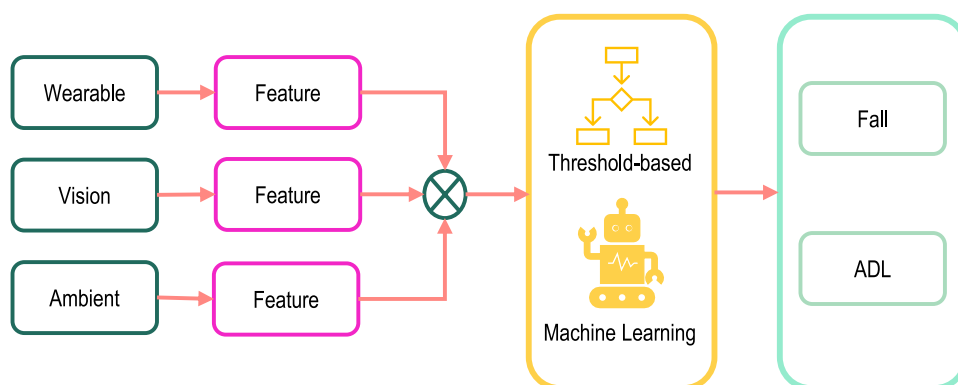


Table 10 Sensors and feature used in multimodal fall detection approaches

Ref	Year	Sensors Used	Features
[230]	2025	Camera, Microphone	Audio features: Short-time Fourier transforms (STFT), Continuous wavelet transforms (CWT); Video features: Visual frames and sequential frame analyses via GoogLeNet and GoogLeNet-LSTM
[231]	2025	Camera, Acc.	Skeletal joint positions, joint motion, time-series acceleration
[232]	2025	Thermal camera, smart-watch, smartphone	Thermal: TRW, PCC, OF; IMU: IMU-Stat, IMU-SHAP/RF, IMU-Norm, IMU-SMOTE
[233]	2025	RGB Camera, Microphone	Video: ResNet-extracted patch tokens with temporal embedding; Audio: log Mel-Spectrogram
[234]	2024	Camera, Microphone	Video features: Key skeletal joints, Spatio-temporal data; Audio features: Log-scaled mel spectrograms
[235]	2023	Accelerometer, Gyroscopes, Camera	Acceleration, Angular velocity, Gray-scale images
[228]	2019	FMCW Radar, Acc, Gyr, Mag	Mean, SD, skewness etc.
[236]	2019	Video Camera	Change of acceleration, deceleration, velocity
[237]	2019	Acc, Camera	Features extracted after segmentation by texture of image
[223]	2018	Acc, photoelectric sensors.	Absolute acceleration
[238]	2018	Camera, audio sensor	N/A
[239]	2017	Surveillance Camera, Acc	Change of human head, log-polar histogram
[240]	2016	Acc x-IMU, camera	Features of non zero pixel after segmentation
[224]	2016	IMU, Kinect v2, Acc	Spine Base Joint location, Acceleration Magnitude, orientation angle, torso angle etc.
[227]	2016	Acc and camera	EO Histogram and GLBP based features
[241]	2015	Kinect, Camera, depth sensor, Acc	Ratio of head floor distance to height, person area in the image proportion to bounding box and blob ratio.
[225]	2015	Kinect, Acc	Sum Vector Acceleration
[229]	2015	pressure sensors, Acc, Gyr, motion sensor	feature vector from CNN
[242]	2014	Kinect, Acc	N/A
[243]	2013	Kinect, Acc	Fall energy image
[244]	2013	Acc, Speech recorder, Camera	tilt angle and signal magnitude vector
[226]	2012	Acc, Color Cameras, IR Camera	visual spectrum and infrared spectrum
[245]	2011	Camera, Acc	bounding box and time sequence motion feature
[246]	2009	Overhead Camera, Acc	Bounding box coordinates are used as feature.

Legend: Acc—Accelerometer; Gyr—Gyroscope; Mag—Magnetometer; IMU—Inertial Measurement Unit; IR—Infrared; SD—Standard Deviation; EO—Edge Orientation; GLBP—Gradient Local Binary Patterns

Table 11 Publicly available datasets on multimodal sensor-based fall detection approaches

Dataset	Simulated/Real	Modality	Indoor / Outdoor	Age	Health	No. of Fall
UP Fall	Simulated	Infrared, Camera, Acc. Gyr, EEG, Light	Indoor	18-24	Healthy	5
CMDFall	Simulated	Kinect sensors, Acc	Indoor	21-40	Healthy	8
FALL-UP	Simulated	Infrared, Camera, Acc. Gyr, EEG	Indoor	NA	NA	5

Legend: Acc.—Accelerometer; Gyr.—Gyroscope

Machine Learning-Based Approaches

Traditional Machine learning (ML) and deep learning (DL) techniques have significantly advanced multimodal Fall Detection Systems (FDS) by enabling the integration of diverse sensor modalities.

Wearable and Ambient Sensor Fusion

Wearable sensors, such as accelerometers and gyroscopes, have been effectively combined with radar systems to improve fall detection accuracy. Li et al. [228] implemented a Bi-LSTM neural network to fuse Frequency-Modulated

Continuous Wave (FMCW) radar data with wearable sensor inputs. The combination of soft feature-level fusion and hard decision-level fusion provided robust performance for identifying falls and distinguishing them from daily activities. Similarly, Wang et al. [229] utilized wearable sensors alongside pressure sensors in a multi-source CNN ensemble, standardizing and compressing sensor data into feature vectors. This system demonstrated versatility by accurately detecting multiple fall types and Activities of Daily Living (ADL).

Wearable and Vision Sensor Fusion

The integration of wearable sensors with vision modalities enhances spatial and temporal feature representation. Tasoulis et al. [245] and Doukas et al. [246] explored systems combining accelerometers with overhead cameras. These systems employed change detection algorithms like Cumulative Sum Algorithm (CUSUM) and adaptive background modeling to track motion and detect falls, using SVM classifiers for final decision-making. Similarly, Turkane et al. [237] fused accelerometer data with image segmentation techniques, employing Gray-level Co-occurrence Matrix (GLCM) features and SVM classifiers for precise fall identification. Qi et al. [235] proposed FL-FD, a federated learning-based system that combined Gramian Angular Field images from wearable sensors with camera data. Kwolek et al. [243] fused accelerometric data with depth images, extracting fall energy images to improve detection rates and reduce false alarms. Shin et al. [231] introduced a multi-modal fall detection system combining skeleton and accelerometer data using a dual-stream architecture consisting of a graph-based spatial-temporal convolutional attention network (GSTCAN) and a Bi-LSTM with channel attention mechanism for robust feature extraction. Lupión et al. [232] designed a privacy-preserving, edge-based fall detection and alert management framework. This framework fuses thermal camera streams and wearable IMU signals. It uses dedicated CNN/ConvLSTM branches and 1D-CNN/LSTM modules. Motion is integrated through optical flow and feature-selected inertial data. Additionally, the system includes a conversational Alexa-based post-fall interview.

Audio and Visual Data Fusion

Multimodal systems incorporating audio and visual data demonstrate the potential for enhanced fall detection in diverse environments. Jiao et al. [234] combined YOLOv7-Pose and two-stream ST-GCN for video analysis with MobileNetV2 for audio classification. By using decision fusion techniques like Dempster-Shafer theory, their system achieved high sensitivity and robustness even in challenging

conditions. Similarly, Geertsema et al. [236] analyzed synchronized audio and video data, extracting acceleration and velocity changes from video sequences and employing SVM classifiers to detect falls, underscoring the benefits of integrating complementary data streams. Dibble and Bazzocchi [230] introduced the BiMP dataset, emphasizing synchronized audio-visual fall data to support privacy-conscious FDS development. By employing GoogLeNet for video and STFTs and CWTs for audio, their approach demonstrated high accuracy while enabling privacy-preserving mechanisms. Ambient and robotic systems leverage environmental sensors and autonomous agents for fall detection. Ciabattini et al. [238] integrated wearable devices with a mobile robot equipped with audio and video sensors. By employing MobileNet and audio recognition, their system enhanced detection reliability through multimodal verification. Zhang et al. [233] proposed Fall-Mamba, a multi-modal fall-detection system that fuses video and audio via a Cross-Attention module. This proposed system models temporal dynamics with a bidirectional Mamba SSM augmented by multihead temporal attention, and injects Frame Masking during training to simulate real-world data loss.

Performance Analysis

Different studies used different performance metrics to measure the performance of their methods. However, we have found that accuracy, sensitivity, and specificity are the most common performance metric. Besides these, false-positive and the number of false alarms are also used to show the performance. The following subsections demonstrate the performance of the studied articles.

Performance of Wearable Sensor-based Fall Detection Approaches

We have analysed the performance of wearable sensor-based FDSs using datasets and algorithms in this section. Table 12 presents the datasets, obtained results, and algorithms implemented in wearable sensor-based FDSs.

Chen et al. [104] proposed a wearable fall detection system embedded with a waist-worn accelerometer and utilised the area under SMC. Experiment results showed that it achieved high reliability and sensitivity to distinguish between falls and ADL using a simulated dataset. Wu and Xue [107] developed a pre-impact threshold-based portable fall detection system using a tri-axial accelerometer and angular rate sensor utilising inertial frame velocity magnitude. The proposed algorithm can detect fall events 70 ms before the impact of the fall and achieve 100% TPR with no false alarm in the case of an individual person-specific

Table 12 Dataset and algorithm used for fall detection using wearable sensor-based approaches

Ref.	Algorithm	Dataset	Samp. Rate	Result
[78]	Threshold	TSTFB4FD	100 Hz	Acc.- 97.1% Spe.- 97.8% Sen. - 96.5%
[100]		Self	25 Hz.	FPR - 0.0079 FPs/day
[104]		Self	200 Hz	High reliability and Sen.
[47]		Self	N/A	At foot, Sen. 100% and Spe. 90%
[52]		Self	1000 Hz	Spe. 100%
[79]		Smartphone Dataset	50 Hz	Sen. 0.979, Spe. 0.967
[53]		N/A	N/A	Sen. 90% and Spe. 86.7%
[96]		Self	N/A	Detection rate of Falling From Standing 92.15%
[80]		Self	N/A	Indoor Acc. 93.78% and Outdoor Acc. 89.8%
[86]		Self	50 Hz	Acc. - 99.69% Sen. - 100% Spe. - 98.89%
[90]		Self	50 Hz	average Acc., Sen. and Spe. is 100%
[87]		Self	100 Hz	Acc. - 90%
[49]		Self	N/A	Not specified
[92]		Self	25 Hz	Acc. - 94.97%
[107]		Self	50 Hz	TPR - 100% , TNR- 100% , Lead time 70 ms
[50]		Self	50 Hz	Sen. 91% false alarm rate 0.1149/hour
[54]		Self	50 Hz	Lead time 157 ms, Sen. 100%
[103]		Self	100 Hz	Not specified
[95]		Self	100Hz	Sen. 100% Spe. 95.65% and RT 255 ms
[99]		Self	100 Hz	Sen.- 100% Spe. - 95.25%
[51]		Self	6,50,25 Hz	Sen. - $91.5 \pm 4.6\%$ Spe. - $96.5 \pm 1.5\%$
[73]		Self	100 Hz	Threshold (Acc. 91.1% Sen. 95.8%Spe. 86.5%;ML(Acc. 99.0%Sen. 100%Spe. 97.9%
[88]	SVM	Self	200 Hz	FPR reduced 50% Track within .39m
[48]		Self	N/A	Not specified
[46]		Sisfall 2	25 Hz	Acc. - 99.4%
[93]		Self	125 Hz	Sen. 95% Spe. 90% LT 0.0625-0.1875s,WS 1s
[72]		SisFall	40 Hz	Acc. - 99.94% Sen. - 99.05% Spe. - 99.95%
[105]		Self	160 Hz	Sen. 100% Spe. 100% average LT of 203 ms
[91]		Self	160 Hz	Spe. - 99.5% Sen. - 97%
[106]		Self	100 Hz	Not specified
[85]		Self	100 Hz	Acc. 97.17% Sen. 97% Spe. 97.50%
[75]		Self	50 Hz	Average Sen. up to 95.8% and Spe. upto 99.2%
[109]	KNN	Self	512 Hz	ratio of correctness 95%
[70]		SisFall	200 Hz	KNN- Acc. 99.80% RF Acc. 96.82%
[247]		WS Dataset	25 Hz	Waist region Sen. - 99.96% by using the k-NN
[98]	HMM	Self	100 Hz	Lead time 200–400 ms, Acc. - 100%
[82]		FARSEEING	20 Hz	PPV - 0.981, Sen. - 0.99
[97]		Self	100 Hz	Precision - 96% , Recall - 95%
[94]	DT	Self	50 Hz	Recall 93.1% Precision 95.9% and F-value 94.5%
[83]		Self	50 Hz	Saving power - 75.3% Acc. - 92.9%
[67]	KNN,SVM,DT,EC, MP,OMP	Self	50Hz	Max Acc. 99.8%
[77]	KNN,SVM,NB,DT	SisFall	200,3 Hz	21.3 Hz is enough for Acc. 97%
[81]	FMNN,GFDA, GKSVN,Fuzzy	Self	1024 Hz	Sen. - 98.70% and Spe. - 98.59% using GK-FDA
[69]	GMM	HRIMU	50 Hz	Sen. 93.09% Spe. 89.58% Acc. 92.22%
[84]	Grey Prediction	N/A	N/A	Fast output response
[102]	LDA	Self	120 Hz	Sen. 96% Spe. 98%
[101]	LR	Self	N/A	Correctness 96.5%
[89]	MB,RF,RoF,AB,RB	Self	1000 Hz	Sen. 81.8% Spe. 72.7%
[45]	N/A	N/A	N/A	Not specified
[108]	One class SVM	Self	512 Hz	Correctness 96.7%
[76]	RF, SVM, KNN	SisFall	200 Hz	RF Acc. 99.42% Sen. 94.42%Spe. 99.63%
[68]	SVM,kNN,CART, NB	SisFall	128	Sliding windows Acc. 99.66%impact-defined windows Acc. 99.78% in KNN
[71]	TSAM,SVM,KNN, RF	Self	20 HZ	Acc. 93.3%Sen. 93.3%Spe. 98.3%

Table 12 (continued)

Ref.	Algorithm	Dataset	Samp. Rate	Result
[63]	BiLSTM+ RNN	SisFall, UP-Fall	N/A	Acc. 97.21% (SisFall), 97.41% (UP-Fall); Sen.: 96.97%, 99.77%; Spe: 93.18%, 91.45%
[60]	Multi-Branch Aggregation with Deep Residual Network	UPFall, SisFall, UMAFall	100 Hz for the UpFall dataset, 200 Hz for the SisFall dataset, and 20 Hz for the UMAFall dataset	Acc. 96.16%, F1-score: 99.12% on SisFall, UpFall, UMAFall datasets
[62]	CNN + BiLSTM + Transformer Encoder	Self	200 Hz	Best (shank): F1 = 0.97, Balanced Acc.= 0.98; Others: Foot: F1 = 0.88, Hand: F1 = 0.85
[61]	RF, SVM, KNN, ANN, GB	Erciyes, UMAFall	20 Hz and 25 Hz	Erciyes: Acc 98.4%, SE 98.9%, SP 96.7% UMA-Fall: Acc 95.5%, SE 98.6%, SP 93.7%
[56]	CNN + Channel Attention, BiLSTM + Temporal Attention	KFall, SisFall	100 Hz, 50 Hz	Acc. 97.93% (KFall), 98.99% (SisFall); Pre.: 97.77–100%; Rec.: 95.67–97.96%
[59]	CNN	SisFall, FALLAIID, KFall	25 hz	Rule-based: F1 = 82.94–89.8%; CNN: F1 = 86.72–95.09%
[57]	CNN, RNN, LSTM, GRU, CNN-LSTM, RNN-LSTM, GRU-LSTM	Self	N/A	Acc. 97%, Rec.: 98%, Pre: 98%, F1: 0.98
[58]	ResNet	Self	125 Hz	Fall detection Acc. 94%, Overall classification acc (12 actions): 85.61%

Legend: Acc.– Accuracy; Sen.– Sensitivity; Spe.– Specificity; TPR– True Positive Rate; TNR– True Negative Rate; FPR– False Positive Rate; FNR– False Negative Rate; SVM– Support Vector Machine; KNN– K Nearest Neighbour; RF– Random Forest; DT– Decision Tree; LR– Logistic Regression; LSM– Least Square Method; LDA– Linear Discriminant Analysis; FMMNN–Fuzzy Min-Max Neural Network; GKFDA–Gaussian Kernel Fisher Linear Discriminant Analysis; GKSVM–Gaussian Kernel SVM; HMM– Hidden Markov Model; OCSVM– One Class SVM; DTW– Dynamic Time Wrapping; GMM– Gaussian Mixture Model; MB– MultiBoost; RoF– Rotation Forest; AB– Adaboost; RB– RusBoost; CART– Classification And Regression Tree; MP– Matching Pursuit; OMP– Orthogonal Matching Pursuit; TSAM– Temporal Signal Angle Measurement; NB– Naïve Bayes; PPV– Positive Predictive Value

threshold value. Basili et al. [47] combined a wearable accelerometer with a smartphone and proposed a threshold-based fall detection algorithm to distinguish between falls and other normal activities in daily life. It is reported that the best result obtain while device was placed at the foot or ankle compared to the arm having maximum sensitivity, specificity and accuracy of 100%, 90%, and 93.4%, respectively. Wang et al. [50] presented a low-energy consumption fall detection system using a tri-axial accelerometer and pressure sensor to balance the performance of the system and the false alarm rate. Experiment results demonstrated that the proposed system achieved sensitivity of 91% and generates 0.1149 false alarms per hour with increased battery life. Lu et al. [51] introduced a fall detection system using a barometric pressure sensor which was triggered by SPM to reduce energy consumption and increased the battery life of the barometer. The proposed system achieved maximum sensitivity and specificity of 91.9% and 95.5%, respectively by using the Kalman filter processing method with pre-defined thresholds having low false alarms per hour. Guo et al. [52] proposed a threshold-based fall detection algorithm incorporating an accelerometer with gyroscope and obtained specificity of about 100% during the experiment.

Nari et al. [53] combined an accelerometer with a gyroscope to develop a wearable fall detection system and

proposed a threshold-based algorithm that can detect fall events with a sensitivity of 90% and specificity of 86.7%. Sabatini et al. implemented a fall detection system to identify the post and prior impact of falls using a barometric altimeter, gyroscope, and accelerometer. Using a threshold-based algorithm, the proposed system achieved a sensitivity of 100% and specificity of 100% [54]. Pierleoni et al. [86] incorporated an AHRS with a barometer to develop a threshold-based wearable fall detection system and found an accuracy of about 100%. A combination of accelerometer, magnetometer and gyroscope sensors was used to implement a wearable fall detector and proposed a threshold-based algorithm to detect falls by Pierleoni et al. [90] Experiment results showed that the proposed system achieved an average accuracy of 90% and sensitivity of 80.74 %.

Montanini et al. [78] proposed a footwear-based fall detection system consisting of three force-sensing resistors and an accelerometer. The proposed system achieved an accuracy of 97.1% using the TST FB4FD fall detection dataset.

Qu et al. [87] proposed a threshold-based algorithm to evaluate a wearable fall detection system embedded with an accelerometer. The proposed algorithm can detect fall events with an accuracy of above 90%. Sucerquia et al. [46] proposed a real-time robust wearable fall detector consisting of an accelerometer to classify between falls and ADL.

Using the Sisfall dataset, the proposed system achieved 99.4% accuracy to detect fall events. Liu and Lockhart [95] carried out a study to develop a wearable IMU-based fall detection system and proposed a threshold-based fall detection algorithm using a trunk angle and trunk angular velocity. The proposed algorithm could detect a fall on average 255 ms response time and achieved 100% sensitivity and 95.65% specificity. Liu and Lockhart [99] developed an integrative ambulatory measurement framework with an automatic person-specific threshold calibration function to detect human falls. The experiment result showed that the proposed system achieved a sensitivity of 100% with a lower response time. Ozacan et al. [96] proposed a wearable camera-based fall detection system using optical flow image and HOG feature descriptor to detect fall events. The fall detection accuracy was 92.15% during experiments with a simulated dataset. Ozcan et al. [80] proposed a threshold-based fall detection system that automatically computes the threshold using relative entropy distance measure. The proposed model achieved a maximum of 93.78% and 89.8% accuracy in indoor and outdoor environments respectively.

Aziz et al. [93] investigated the result of lead time and window size on a fall detection system using a machine learning-based SVM classifier and developed an inertial sensor-based pre-impact fall detection system. It was found that the proposed model achieved a sensitivity of 95% and specificity of 90%. Shibuya et al. [91] developed a WGAS system consisting of an accelerometer and gyroscope sensor to distinguish between fall events and ADLs using an SVM classifier. The proposed system achieved a fall classification accuracy of about 98.8%. Saleh and Jeannes [72] proposed a low computational cost fall detection algorithm using an online feature extraction algorithm and SVM algorithm to classify fall and non-fall events. The proposed algorithm achieved 99.9% accuracy on an open Sisfall dataset with low computational cost. Shi et al. [106] implemented an SVM-based fall detection system with a mobile airbag protection system and found a fall detection accuracy of about 100%. Shan and Yuan [105] carried out a study to develop a pre-impact fall detection system using a tri-axial accelerometer and employed an SVM classifier to distinguish between falls and ADL. The proposed system achieved sensitivity of 100% and could detect fall events 203 ms before the impact of the fall. Liu et al. [77] investigated the effect of sampling rate on the performance of a wearable fall detection system using machine learning models such as KNN, SVM, Naive Bayes, and decision tree. SVM with radial basis function outperformed the other machine learning model and achieved fall detection accuracy of about 98% at a sampling rate of 11.6 Hz.

Shastri et al. [88] proposed a fall detection system consisting of an inertial sensor and radio frequency ToF

transceiver. The system achieved a maximum true positive rate of 99% using supervised SVM with RBF kernel. Quadros et al. [73] used Madgwick's decomposition method to develop a wrist-mounted wearable fall detection system and evaluated the proposed system by comparing results of KNN, LDA, LR, DT, and SVM classifiers. KNN classifier outperformed the other classifiers and showed an accuracy of 99% for fall detection. Zhang et al. [108] presented a wearable fall detection system using a tri-axial accelerometer and NMF algorithm. Using the KNN classifier, the proposed system can detect fall events with a ratio of correctness of 98.4%. He et al. [85] used the KNN algorithm with a Kalman filter to develop a wearable fall detection system consisting of an accelerometer and gyroscope. The proposed system can differentiate between fall events and ADL with an accuracy of about 97.17%.

He et al. [75] developed a fall detection system using an accelerometer and gyroscope and used the KNN algorithm to detect fall events. The experiment result reported that the proposed model could detect fall events with an accuracy of about 95.8%. Liu et al. [68] investigated the effect of window size and segmentation approach on a machine learning-based wearable fall detection system. Using the Sisfall dataset, the system achieved a maximum of 99.78% accuracy using the KNN classifier with an impact-defined window method. Hussain et al. [70] proposed an activity-based fall detection and fall pattern recognition system using a wearable sensor and compared the performances of KNN, SVM, and random forest classifier. Using KNN, the proposed system can detect fall events with a maximum accuracy of about 99.8% and outperformed the other machine learning classifiers. Ozdemir [247] carried out a study to investigate the best sensor location on the subject's body by comparing six machine learning models. The experiment results obtained the highest accuracy of 99.7% with the KNN classifier in the waist region among several combinations of sensors. Kerdjadj et al. [67] employed compressed sensing technology with an accelerometer and gyroscope to classify falls and ADLs. Several machine learning algorithms such as KNN, SVM, DT, and Ensemble methods were used to detect fall events and achieved maximum accuracy of 99.8%. Aziz and Robinovitch [102] investigated the causes of human falls by developing an accelerometer-based wearable fall detection system. Experiment results depicted that the proposed system can successfully classify causes of human falls using the LDA algorithm and achieved a sensitivity of 96% and specificity of 98%. Melillo et al. [89] employed a wearable ECG sensor to develop a fall detection system and calculated HRV from the ECG signal to detect fall events. The proposed system obtained 72% accuracy to identify future fallers among hypertensive patients and achieved a fall classification accuracy of about 77.3% using LDA.

Nho et al. [69] combined an accelerometer and heart rate sensor to develop a user-adaptive fall detection system using a cluster analysis-based GMM model. The system showed a fall detection accuracy of about 92.22% during the experiment. Tong et al. [98] proposed an accelerometer-based fall prediction and detection system that used HMM classifier to distinguish between falls and ADLs and results showed that it can predict an average of 200 to 400 ms prior to the fall event and achieved fall detection accuracy of 100%. Yu et al. [82] used HMM and sensor orientation algorithm to develop a wearable fall detection system which consists of an accelerometer. The proposed model achieved a positive predictive value of 78.6% using the FARSEEING real-world fall dataset. Pham et al. [97] used Wii remotes to develop a fall detection system that can identify thirteen fine-grained activities in real time. An HMM classifier was employed to distinguish between fall and other activities and achieved a precision of about 95%. Wang et al. [83] employed a backward wrapper feature selection algorithm to develop a power-efficient fall detection system and used a binary decision tree classifier to distinguish between fall and non-fall events. The proposed system could detect fall events with a maximum classification error of 7.1%. Luo et al. [94] integrated three accelerometer sensors to develop a wearable fall detection system using a decision tree classifier and achieved a recall of 93.1% for fall classification.

Hussain et al. [76] created wearable fall detection and fall pattern recognition system using a supervised machine learning approach. The system achieved maximum accuracy of 99.45% for fall pattern recognition using the RF algorithm and outperformed the SVM and KNN algorithms. Pereira et al. [101] developed distributed WSN-based fall detection system using the accelerometer sensor and logistic regression classifier that can classify fall events with an accuracy of 98%. Kwon et al. [71] introduced a power-efficient TSAM algorithm to detect fall events using an accelerometer sensor and results were compared with SVM, KNN, and random forest classifiers. The proposed TASM algorithm outperformed the SVM, KNN, and random forest classifiers and achieved an accuracy of 93.3%. Medrano et al. [79] presented a new fall detection system integrating a novelty detector and made a comparison with traditional SVM-based fall detection systems. The proposed system achieved a fall detection sensitivity of about 97.9% using a publicly available smartphone dataset. Xi et al. [81] employed the sEMG sensor to obtain the best feature and classification model for fall detection and activity classification. It was observed that a combination of the Gaussian Kernel Fisher Linear Discriminant Analysis (GKFLDA) classification model with Wilson Amplitude (WAMP) features achieved the highest sensitivity and specificity of above 98% with a low calculation time of 65 ms. The system proposed by

Vyas et al. [55] monitors resultant acceleration and pressure, by applying sustained thresholds and smoothing transient spikes through a damping factor. This approach enhances sensitivity and reduces false positive rates when compared to existing wearable methods. Additionally, real-time Bluetooth alerts ensure prompt notification for caregivers. The wearable sensor and deep learning based system proposed by Waheed et al. [63] achieved 97.21% accuracy with SisFall and 97.41% with UP-Fall and 99.77% sensitivity and 93.18% specificity on SisFall- maintaining performance even under noisy or missing sensor data conditions. Fall-NeXT model by Mekruksavanich and Jitpattanakul [60] outperformed existing methods with an accuracy of 96.16% and F1-score of 99.12% for fall and activity recognition across multiple datasets (SisFall, UpFall, UMAFall). The compact model with high efficiency make it suitable for wearable sensor applications in real-time monitoring. The wearable sensor based system developed by Yhdego et al. [62] demonstrated high performance metrics (F1 score of 0.97 and a balanced accuracy of 0.98). Their approach of integrating multi-window segmentation and a customized focal loss function ensures better fall detection and differentiation from near-falls.

Ruiz-Garcia et al. [61] evaluated their proposed "Care-Fall" system on two public datasets (Erciyes and UMAFall), achieving superior performance with machine learning classifiers using combined accelerometer and gyroscope features. The best configuration reached 98.4% accuracy and 98.9% sensitivity on the Erciyes dataset using Random Forest, and 95.5% accuracy and 98.6% sensitivity on UMAFall using SVM, significantly outperforming traditional threshold-based approaches. The ensemble fall detection model by Sahni et al. [56] demonstrated higher performance than the baseline CNN and BiLSTM models using two different publicly available datasets (98.99% accuracy on SisFall and 97.93% accuracy on KFall dataset). Such performance ensures the effectiveness of the proposed technique of the model. The low-cost fall detection device by De Raeve et al. [59] used a compact BLE 5.1 wearable at the waist to detect falls via SMV thresholds ($>2.5g$) and orientation angle shifts ($>60^\circ$), achieving F1-scores up to 89.8% across multiple datasets. Their rule-based system proved highly energy-efficient and interpretable, while the CNN benchmark reached up to 95.09% F1-score but required more computational resources and training. A et al. [57] achieved the highest performance (97% accuracy and 0.98 F1-score) using a CNN-LSTM with attention layer model on stratified elderly motion data collected in controlled indoor settings. The optimized model was deployed on edge devices like Raspberry Pi 4 and Jetson Nano, validating the system's real-time efficacy and suitability for practical elderly monitoring. The pressure-insensitive insoles developed by Guo

et al. [58] achieved a fall detection accuracy of 94% using ResNet model trained on time-series plantar pressure maps. The model was able to distinguish between 12 actions with 85.61% accuracy and received high user ratings (privacy: 4.13/5, utility: 4.53/5) in a usability study involving volunteers aged 37–85.

Performance Analysis and Design Insights into Wearable Sensor-Based Fall Detection Systems

Wearable fall detection systems benefit from direct motion sensing, enabling high sensitivity to body dynamics through accelerometers, gyroscopes, and pressure sensors [50, 52, 104]. Threshold-based methods are lightweight and energy-efficient but often lack adaptability across individuals, leading to false alarms or missed detections [47, 51]. Machine learning approaches, including SVM and KNN, improve personalization and pattern recognition [72, 85, 93], though their effectiveness can degrade with sensor noise or hardware inconsistency. Deep learning and hybrid models [56, 60, 63] demonstrate robust performance, even under missing or noisy sensor inputs, but often require greater power and memory, limiting use in compact devices. Additionally, sensor placement critically impacts performance; wrist or ankle-mounted devices may miss certain fall types [62, 94]. While wearable systems are unobtrusive and suitable for real-world monitoring, trade-offs exist in accuracy vs. generalizability, energy consumption vs. model complexity, and sensor fusion vs. hardware cost. Continued research is needed on energy-efficient architectures, adaptive thresholds, and multi-sensor fusion for robust and scalable deployment.

Performance of Vision Sensor-Based Fall Detection Approaches

Vision-based fall detection is becoming popular with the enhancement of computation power. This section briefly analyses the obtained performance of different FDSs using vision sensors. The datasets, algorithms and performance of vision-based studied systems are summarised in Table 13. Rajabi and Nahvi [158] developed a video surveillance camera-based fall detection system using a threshold-based approach where GMM was incorporated to extract out moving objects from an image. Experiment results demonstrated that the proposed system achieved fall detection accuracy of about 93.59%. Nguyen et al. [153] proposed a single camera-based fall detection system and applied a threshold on motion and human shape features to distinguish between fall and non-fall events. An experiment was performed on a publicly available dataset named Le2i which gives a maximum accuracy of 100%,

as well as some false alarm, was generated. Albawendi et al. [147] analysed the time motion history image and human shape variation to develop a threshold-based fall detection system and observed fall detection accuracy of about 99.81% during the experiment. Leone et al. [178] incorporated a ToF three-dimensional camera and implemented a threshold-based fall detection system using computer vision technology to detect falls and human postures. The proposed system could detect fall events with a reliability of 97.3%. Auvinet et al. [177] used multiple cameras to develop a vision-based fall detection system and employed an occlusion-resistant algorithm with VVDR to distinguish between fall and non-fall events. The proposed system detected a fall event with a sensitivity of about 99.7% using a publicly available multi-camera fall dataset.

Joshi and Nalbalwar [150] proposed a vision-based fall detection system applying a threshold on moving object features and found fall detection accuracy of about 91.89% using the Le2i dataset. Nghiem et al. [174] used a Kinect depth sensor to develop a fall detection algorithm with the help of a head detection algorithm that was proposed in this work. A simulated dataset was created to evaluate the system which showed that the system could successfully detect 29 falls from 30 fall events.

Edgcomb and Vahid et al. implemented a vision-based fall detection system that can detect fall events from privacy-enhanced video using a background subtraction algorithm. A binary tree classifier was used to classify fall and no fall events and achieved a sensitivity of 91% and specificity of 92% [173]. Nizam et al. [145] included fall risk level identification to develop a user-adaptive fall detection system using a depth sensor. The author proposed a threshold-based fall detection algorithm and achieved a fall detection accuracy of about 88.57% using the URFD dataset. Merrouche and Baha [154] analysed the human shape and motion features to develop a threshold-based fall detection system using a depth sensor and realised fall detection accuracy of 92.98% using the SDUFall dataset. Kroputaponchai and Suvonvorn [179] proposed a vision-based fall detection system analysing motion and human shape features and obtained fall detection accuracy up to 80%. Kothari et al. [140] developed a vision-based fall detection system for indoor environments using three uncalibrated fisheye cameras that can detect fall events with an accuracy of 94%. A fuzzy model introduced in [157] to detect fall and fall patterns from a colour video camera which can detect fall and fall patterns from a set of static and dynamic fall activities with an accuracy of 100%.

Yu et al. [166] employed the OCSVM algorithm to develop a user-specific fall detection system using a single USB camera and found fall detection accuracy of 100% having a false alarm rate of 3% during experiments. Yu et

Table 13 Dataset and Algorithm used in vision-based approaches

Ref.	Algorithm	Dataset	Result
[140]	Threshold	PIROPO	Acc. Above 94%Sen. above 82.29%
[145]		URFD	Acc. 88.57% and Spe. of 82.5%
[147]		Self	Sen. 95.23%Spe. 100% Acc. 99.35 %
[148]		N/A	N/A
[150]		Li2e	Selectivity 88.23%Spe. 92.98%Acc. 91.89%
[153]		Li2e	Recall-Lecture-room 100%
[154]		SDUFall	Acc. 92.98% ;Sen. 93.52%Spe. 90.76%
[158]		Self	Acc. of 93.59% and Sen. of 86.67%
[173]		Self	Sen. (0.91) and Spe. (0.92)
[174]		Self	Acc. 97%
[177]		Self	100% Sen. and Spe. with 4 cameras;
[178]		Self	Reliability of 97.3% Efficiency 80%
[179]		Self	Detection Rate Up to 80%
[137]		SDUFall, URFD, MultiCamera	SDUFall- Acc. 96.04%UR Fall -100% Sen. and 95.00% Spe.
[152]		URFD	Precision 96.8% Recall 98.1% time complexity 0.17s per frame.
[159]	SVM	Self	Sen. 76.1 % PPV-41.2 %
[160]		Self	Prediction Acc. 82.18% at half progress
[163]		Self	Acc. 97.5% and FP -1 for SVM; Localization Acc. 40-55 cm
[164]		Self	Sen. 95.8%;Spe. 100%Acc. 97.9%
[167]		Self	average precision 90.46%
[168]		Self	Detection Rate 95%
[169]		Self	0.984 TPR Mean;0.987 TNR Mean
[171]		Multicam	Error rate 4.6%
[149]		Self	Sen. 96%Spe. 97.6%Precision 96%Acc. 96.9%
[155]		Self	Successfully detect
[181]		Self	Recognition Rate 91.12%
[180]		Self	Recognition Rate 89.99%
[146]		3D CNN with LSTM	Acc. 100%
[138]		CNN	NTU RGB+D
[136]		CNN - VGG-16	URFD, Multicam, FDD
[175]	DAGSVM	Self	97.08% TPR, .8% FNR
[139]		DNN	98.5% Acc.
[161]		EDT	98% 70% and 71% detection of standing, sitting, and lying down falls
[143]		Faster R CNN	URFD
[172]		FFNN	self
[157]		Fuzzy Logic	Self
[151]		GLR- SVM, NN, KNN, DT, NB	URFD, FDD
[176]		GMM	Self
[182]		HMM	Self
[170]		k-centre, KNN, OCSVM, SCMPM	Self
[142]		KNN,SVM,RF,DT	URFD, Le2i
[141]		LBAdaptativeSOM	Self
[144]		LMNN	Self
[156]		NB	WAD, SDUFall
[165]		NN, FSM	Self
[166]	OCSVM	Self	100% TPR, 3% FNR
[162]		VPSO-ELM	SDUFall
[128]		RF	UR Fall Detection, Le2i
[127]		Transformer-based – Uniformer	UP-Fall, UR Fall
			Acc. 89.99%, Sen.: 90.33%, F1-score: 90.02%, FPS: 29.7
			Acc. 99.17%, F1-score: 94.14% (binary, UP-Fall); Acc. 93.17%, F1-score: 93.39% (multiclass, UP-Fall); F1: 94.76% (UR Fall, jointly trained)

Table 13 (continued)

Ref.	Algorithm	Dataset	Result
[129]	4-stream 3D CNN + Human Segmentation + Image Fusion	Le2i	Acc. 99.44%, Sen.: 99.12%, Spe.: 99.12%, Pre.: 99.59%
[124]	High-Resolution Network (HRNet) + Spatio-Temporal Coordinate Attention + 3D/Pseudo-3D Convolution	MCF, Lei2i, NTU RGB+D120	Acc. 99.8% (Le2i), 99.37% (MCF); Sen.: 100%; Spe.: up to 98.23%
[125]	3D Convolutional Autoencoder, Attention U-Net, Multi-modal Network	MUVIM	AUC ROC up to 0.929 (multi-modal), AUC PR up to 0.126 (Orbbec IR + Depth)
[130]	Graph Convolutional Network + Separable Temporal Convolutional Network	URFD, UPFD, UWA3D, NTU-60, NTU-120	UWA3D (cross-view): Acc.: 98.88%, 94.06% AUC
[131]	Improved YOLOv5s + ACB + Spatial Attention	URFD	mAP: 97.2%
[126]	YOLOv7	Self	Acc. 96%, Pre.: 99.1%, Rec: 93.6%, F1-score: 96%, FPS: 30
[123]	YOLOv8	CUCAFall, DiverseFALL10500	mAP: 0.9955 (CUCAFall), 0.950 (DiverseFALL10500); F1-score > 0.998 (CUCAFall)
[122]	YOLOv8s + AlphaPose + Hybrid Fall Detection Algorithm (HFDA)	MPFDD, Le2i	Acc. 77.5%, F1: 0.774, Pre.: 78.7%, Rec: 76.1%, FPS: 19.8
[121]	Temporal Convolutional Network + Transformer Encoder	Up-fall, LE2i, GMDCSA-21	Window-wise Acc. UP-Fall front 99.58%, side 98.75 %; Le2i 97.01 %; GMDCSA-24 92.99
[120]	ESNet-L backbone + H-Swish activations + RepConv modules + C2 detection head	E-FPDS, URFD, UPFD	mAP@0.5: 78.4% on E-FPDS, 99.2% on URFD; 99.5% on UPFD (single-dataset tests)

al. [175] combined the human posture and floor information to detect human fall events using DAGSVM and obtained a fall detection rate of 97.08% with 0.8% false alarm. Josemans et al. [169] combined skeleton-based and bounding ellipse-based fall classification methods to distinguish between fall and non-fall events using a depth sensor and RGB camera. An SVM classifier was then used and found TPR of 98.4% for fall detection using a simulated dataset. Bian et al. [164] developed a depth camera-based fall detection system and proposed an SVM classifier to detect fall events. The system could detect fall events with an accuracy of 97.6%. Mirmahboub et al. [171] analysed the silhouette area to develop a single camera-based fall detection system and employed an SVM classifier to distinguish between fall and ADLs. Using a multi-camera fall dataset, the proposed system could detect fall events with an accuracy of 94.18%. Makantasis and Doulamis [168] used the idea of 3D physical space and utilising a single monocular camera with an SVM classifier to develop a fall detection system. It is reported that the proposed model achieved a fall detection rate of at least 95% in a complex background. A single RGB camera was used by Harrou et al. in [151] that combined GLR and SVM in order to efficiently distinguish between fall and fall-like activities. Experiment results showed that the proposed system can detect fall events with an accuracy of 96.66% using two publicly available datasets. Wang et al. [167] proposed a multi-camera-based fall detection system using an SVM classifier to detect fall events and achieved an average precision of 90.46%. Williams et al. [163] employed multiple camera networks to detect and locate fall events using the SVM algorithm and found a fall

detection accuracy of 97.5% with one false alarm. Alaoui et al. [142] used the key skeleton joint of humans with a spatiotemporal method to develop a fall detection system and compared the results of the system using RF, SVM, decision tree, and KNN classifiers. Experiment results showed that the SVM classifier outperformed the other classifiers and achieved fall detection accuracy of 98.5% using URFD and Le2i dataset. Alazrai et al. [160] analysed the MPGD feature to develop a fall prediction system using an SVM classifier and found that the proposed system can predict fall with an average accuracy of 82.18% at half progress level. Debarde et al. [159] proposed a hybrid fall detection model combining a background subtraction algorithm and particle filter with an SVM classifier. The proposed model could detect fall events with a sensitivity of 76.1%.

Yu et al. [170] developed a vision-based fall detection system using one class boundary-based algorithms to detect fall events. The results showed that the SCMPM boundary classifier gives better performance with TPR of 100%. Ma et al. [162] used the VPSO-ELM algorithm with CSS features extracted from a human silhouette to detect falls and other ADLs. The proposed algorithm achieved a fall detection accuracy of 86.83% using the SDUFall dataset. Akagündüz et al. [156] employed an SOV descriptor to detect fall events using a depth sensor and Naïve Bayes classifier. The results showed that the proposed model could detect fall events with an accuracy of 91.89% using the SDUFall dataset.

Zhao et al. [144] proposed a fall detection system to detect fall from bed using a depth sensor and employed the LMNN-based KNN algorithm for the classification of fall and ADLs. It was found that the proposed system could

detect fall events with an accuracy of 92.3%. de Miguel et al. [149] described a camera-based fall detection system using a combination of optical flow algorithm, Kalman filter and KNN classifier to detect falls. Experiment results on a self-made dataset showed that the proposed system achieved a fall detection rate of 96.9%. Stone et al. [161] used an ensemble of decision tree classifiers to develop a Kinect camera-based fall detection system and found a maximum fall detection rate of 98% for standing falls. Rougier et al. [176] used the GMM algorithm for a fall detection system to differentiate real-world falls from normal and lure activities in the home environment and results showed that it achieved an equal error rate of 3.8%.

Humenberger et al. [172] proposed a bio-inspired fall detection system using a stereo vision sensor and neural network to identify fall events. Experiment results denoted that the proposed system achieved a fall detection rate of 96%. Alhimale et al. [165] developed a neural network-based fall detection system using a video camera and achieved sensitivity and specificity of at least 90% for fall detection. Foroughi et al. [165] utilised the Integrated Time Motion Image (ITMI) and Eigenspace methods to obtain a fall detector that used neural networks for fall detection. The result showed that the proposed model realised an accuracy of 89.99%. An intelligent vision-based fall detection system was developed by Foroughi et al. [181] and proposed an MLP neural network to classify human falls. The result showed an accuracy of 91.12% for fall detection. Shen et al. [139] incorporated a DNN to develop a vision-based fall detection system using a cloud environment and realised fall detection accuracy of 100% during testing.

Min et al. [143] presented an R-CNN-based fall detection approach to detect falls on indoor furniture and achieved fall detection accuracy of about 95.50% using the URFall dataset.

Zhang et al. [137] utilised CNN to propose a TDRD-based fall detection system and incorporated an SVM classifier to distinguish between falls and ADL. The proposed system achieved 96.04% accuracy on the SDUFall dataset. Tsai and Hsu [138] analysed the skeleton information to develop a deep CNN-based fall detection system and found fall detection accuracy of about 99.2% on the NTU RGB+D publicly available dataset. Núñez-Marcos et al. [136] incorporated a transfer learning approach to detect fall events using CNN classifier. Three publicly available datasets were used to evaluate the performance of the proposed system and achieved a sensitivity of 100%, 85%, and 97% for URFD, Multicam, and FDD datasets respectively. Lu et al. [146] combined a 3D CNN with the LSTM model to present a deep learning-based fall detection system and obtained a fall detection accuracy of 99.36% and 99.27% for URFD and FDD dataset, respectively. Feng et al. [152] utilised the

MHI and HOG features to develop a vision-based fall detection system using CNN with an SVM classifier. Experiment results showed that the proposed model achieved a precision of 96.8% on the URFall dataset.

The hybrid fall detection system by Wang and Deng [128] achieved 89.99% accuracy and 29.7 FPS on a CPU using only a 480p webcam. The model demonstrated robust sensitivity (90.33%) and efficient runtime, ensuring its applicability for real-time elderly care monitoring. On the UP-Fall dataset, the model by Núñez-Marcos and Arganda-Carreras [127] attained 96.67% accuracy with an F1-score of 82.24% in a multiclass evaluation, and in a binary setting achieved 99.17% accuracy and a 94.14% F1-score. Joint fine-tuning on UP-Fall and UR Fall showed strong generalization, with UR Fall performance rising to an F1-score of 94.76% when oversampling was applied. Alanazi et al. [124, 129] achieved 99.8% accuracy and 100% sensitivity on the Le2i dataset. Their pseudo-3D convolutional strategy enabled fast and accurate skeletal feature extraction across frames, outperforming other state-of-the-art models in complex multi-angle datasets like MCF and NTU RGB+D 120. Denkovski et al. [125] applied Temporal Shift to 3DCAE and Attention U-Net architectures, achieving AUC ROC scores up to 0.929 and AUC PR up to 0.126 on the MUVIM dataset. Zahan et al. [130] demonstrated that their proposed S DFA model achieves up to 98.88% accuracy and 94.06% AUC on the UWA3D dataset under cross-view evaluation, while maintaining low computational complexity (1.15 GFLOPS) and faster inference (0.27ms improvement), validating its efficiency and robustness across multiple public datasets and evaluation settings. The vision based fall detection system by Chen et al. [131] achieved a 97.2% mean average precision (mAP) on the URFD dataset. Their model outperformed standard YOLOv5s by 3.5%, showing significant improvement in detecting both fall and non-fall activities across diverse indoor settings and lighting conditions. Elwaly et al. [126] trained a YOLOv7-based fall detection model on a custom 10K-image dataset and optimized it via quantization and pruning for deployment on Jetson Nano. The system achieved 96% accuracy and 99.1% precision, processing 30 FPS in real-time while maintaining consistent detection performance at distances between 1.85–2.2 meters. The proposed model by Sanjalawe et al. [123] aims to detect human in video frames and analyze their change of posture with the implementation of YOLOv8 Time-Space Transformers. The model achieved high accuracy (mAP of 0.9955 on CUCAFall and 0.950 on DiverseFALL10500), with F1-scores exceeding 0.998. Liu et al. [122] implemented a hybrid fall detection model integrating YOLOv8s and AlphaPose, that significantly improved fall detection speed and reduced model size while ensuring robust performance in

multi-person environments. The proposed model achieved 77.5% accuracy and 0.774 F1 score on the MPFDD dataset. Using three-fold cross-validation on UP-Fall, Le2i, and GMDCSA-24 datasets, TCNTE achieved window-wise accuracies of 99.58% (UP-Fall front), 98.75% (UP-Fall side), 97.01% (Le2i), and 92.99% (GMDCSA-24) [121]. Deployed on an NVIDIA Jetson Orin NX, the full pipeline sustained 19fps for single-person and 17fps for two-person scenarios, demonstrating both high detection performance and real-time edge applicability. The YOLO-fall model by Zhang et al. [120] attains 78.4% mAP@0.5 on the E-FPDS public dataset with just 2.45M parameters and 12.2 GFLOPs, while achieving 99.2% and 99.5% mAP on URFD and UPFD respectively. Under cross-dataset tests, it maintains above-50% mAP, and on expanded/merged datasets reaches up to 95.8%. The model also demonstrated its real-time capability for mobile and small-edge devices (runs at 20.6FPS on a Jetson Nano and 49.8FPS on a Jetson Xavier NX).

Performance Analysis and Design Insights into Vision-Based Fall Detection Systems

Vision-based fall detection systems offer high detection accuracy by leveraging rich spatial and temporal features, especially with deep learning models such as CNNs and Transformers [124, 127, 129, 130]. However, traditional threshold-based and ML methods [153, 156, 158, 162] often lack robustness in complex, cluttered, or low-light environments. Deep learning improves generalization but introduces computational overhead, requiring large datasets and limiting deployment on edge devices. Real-time YOLO-based systems [126, 131] show promise but struggle in occluded scenes. A persistent limitation across all vision-based methods is privacy, particularly in sensitive indoor settings, despite efforts using silhouettes or depth sensing [144, 176]. Overall, while vision approaches excel in detailed fall characterization, trade-offs in privacy, computational cost, and real-world generalizability remain key barriers.

Performance of Ambient Sensor-Based Fall Detection Approaches

This section describes the fall detection performance of studied ambient sensor-based fall detection articles. Table 14 summarised the performance, datasets and algorithms in ambient sensor-based studies.

Li et al. [215] proposed single-channel BSS with NMF and multi-channel BSS with joint NMF source separation methods to improve fall detection accuracy in interference and noisy environment and found 80% accuracy at high

interference with moderate background noise. However, the proposed algorithms showed worse performance at low interference and a small noise level.

Paolini et al. [208] employed RFID reader to locate the indoor position and detect the fall of a person using the monopulse radar technique. The proposed system generates zero false positives and zero false negatives during the experiment. Redmond et al. [214] conducted a pilot study to evaluate a PIR and load sensor-based unobtrusive fall detection system in real-world settings at nighttime. The system could detect fall events with 0.53 false alarms per day. Wang et al. [213] introduced a device-free unobtrusive fall detector using channel state information named as “WiFall” and achieved fall detection accuracy of about 90% using one-class SVM. Random forest algorithm improves the performance of the proposed system with an average accuracy of 94% and a false alarm rate of 13%. Min et al. [210] implemented an SVM-based fall detection system using Kinect sensors. The proposed system achieved fall detection accuracy of about 92.05% using the telecommunication systems team (TST) fall detection dataset. Popescu et al. developed an acoustic sensor-based fall detection system using KNN. The proposed system achieved a sensitivity of 100% with five false alarms generated per hour [219]. Ariani et al. [216] developed a fall detection system consisting of PIR, microwave, and pressure mat sensors and incorporating a decision tree classifier to distinguish between falls and daily activities with multiple people. It is reported that the proposed model can detect fall events with an accuracy of 89.33%, sensitivity of 100%, and 77.14% specificity.

Rimminen et al. [217] presented a fall detection system using floor sensors and NFI technology. Bayesian filter and Markov chain model were employed to identify fall events and achieved a sensitivity of 91% and specificity of 91%. Zigel et al. [218] employed floor vibration and sound sensors to detect fall events using a simulated fall dataset. Using the Bayesian decision rule classifier, the proposed method could detect fall events with sensitivity and specificity of 97.5% and 98.6% respectively. Wu et al. [212] implemented a fall detection system using radar-based doppler time-frequency features and sparse Bayesian classifier. It is found that the proposed system can successfully detect fall events with one misclassification during experiments. Rafferty et al. [209] used a thermal vision sensor to develop a fall detection system and applied logical and machine learning processes to identify humans falls. The proposed system achieved a fall detection accuracy of about 76% using a logical process. But CNN showed better accuracy of 80% than the logical process.

The proposed fall detection system by Newaz and Hanada [192] can differentiate between fall and normal activities using EMD values (low EMD values indicating normal

Table 14 Dataset and algorithm used in ambient sensor-based fall detection approaches

Ref.	Algorithm	Dataset	Result
[208]	Threshold	Self	No FPR and FNR
[211]		Self	N/A
[214]		Self	0.53 false alarm per day
[215]		TigerPlace	80% acc. When noise was high
[209]		Self	Detection rate 76%80% and 36% for Logical process, scene analysis and composite scene analysis respectively.
[218]	Bayesian Decision Rule	Self	Sen. 90%
[217]		Self	Sen. 91% ,Spe. 91%
[212]		Self	One Misclassification
[216]		Self	Sen. 100.00%Se. 77.14% Acc. 89.33%
[219]	heuristic DT	Self	Sen. 100% 5 False alarm/hour
[210]	KNN	Self	Sen. 100% 5 False alarm/hour
	SVM	TST Fall V2.	Acc. 92.5%
[213]	SVM, RF	Self	SVM Acc. 90% RF Acc. 94%
[200]	SVM	Self	Sen. 0.93; Spe. 0.90; F1: 0.92
[194]	Weighted fusion of ERF, SSE, KLC, RF	Lo-Fall	Acc: 93.3%, Pre. 91.9%, Rec 95.9%, F1 score 93.9%
[195]	ResNet	Self	Overall recognition Acc. 96.77%; per-gesture accuracy $\geq 92\%$; average prediction time: 36.84 ms; user- and location-aware tests $> 90\%$ and $> 94\%$ accuracy respectively
[196]	HMM, EoA-based reconfirmation	G-Fall	Pre.: 95.74% (anti-static floor), 97.36% (concrete floor); False alarm: 5.30% to nearly 0% with EoA reconfirmation
[197]	LSTM	Self	Acc. up to 99.5%, Validation: 98.5%, Train: 99.3%
[198]	RF, AdaBoost, SVM, Decision Tree, NB	Self	Best result with 3 PIR sensors: AB Acc. 99%, RF: 98% 2 PIR: AB 98%, RF 96%; 1 PIR: AB 86%, SVM 89%
[199]	Self-supervised Variational Autoencoder + Online update	Self	Acc. 98.3% (lab test), 94.1% (real-life), FPR: 1.4% (after 3-day continuous use)
[193]	SVM, Shallow Neural Network	Self	Acc. 93.7% (SVM), 94.5% (NN), Real-time: 80% accuracy, avg. response time: 15s
[202]	LSTM-CNN, SVM, KNN, C4.5 DT, LSTM, Bi-LSTM	Self, ESC-50	Average F1-score of 0.7705 (± 0.2295) and overall classification acc. around 77% across noise levels SNR0–18dB
[203]	HMM + SVD + K-means clustering	Self	Acc. 92.6%, Recall: 90.4%, Pre.: 92.6% in default environment
[201]	Transformer	Self	Acc. 86.73%, F1-score: 62.34%

Legend: Ref.– Reference; Sen.– Sensitivity; Spe.– Specificity; Acc.– Accuracy; SVM– Support Vector Machine; RF– Random Forest; DT– Decision Tree; FPR– False Positive Rate; FNR– False Negative Rate; KNN– K Nearest Neighbour

movements(e.g., 0.726–1.810) and higher EMD values (up to 23.281) indicates fall events). Using data from 18 volunteers performing walking, squatting, and forward/backward falls, Yang et al. [200] tuned an SVM on the fused feature set, achieving a sensitivity of 0.93, specificity of 0.90, and F1-score of 0.92 under cross-validation, outperforming other supervised ML classifiers, and demonstrated a robust, low-compute fall detection suitable for indoor smart-home deployment. Evaluated on 1,500 samples from 20 volunteers, LoFall by Zhang et al. [194] achieved 93.3% overall accuracy, 91.9% precision, 95.9% recall, and a 93.9% F1-score across through-wall distances of 2–10m. It peaked at 97.6% F1 at 2m and maintained 88.3% accuracy at 10m, demonstrating robust, real-time, privacy-preserving fall detection for smart home environments. Zhao et al. [195] applied a 34-layer ResNet to a custom RFID dataset with

five fall types and five daily routines, achieving 96.77% accuracy, $\geq 92\%$ recognition for each action, 36.84 ms inference time, and robust performance across users ($\geq 90\%$) and tag placements ($\geq 94\%$), demonstrating its effectiveness for real-time, non-intrusive indoor fall monitoring. Wu et al. [196] trained an HMM template on vibration features from 24 subjects under leave-one-user-out evaluation, achieving up to 97.36% detection precision and reducing false alarms from 5.30% to nearly 0% on concrete floors through the EoA reconfirmation mechanism, demonstrating reliable, real-time fall detection in typical indoor settings. Kittiyapunya et al. [197] used LSTM model for fall classification using z-axis point clouds and Doppler velocity as input data, achieving up to 99.5% classification accuracy across five real-world environments. The system effectively suppressed false positives and demonstrated real-time detection capability while

minimizing computational complexity and overfitting risks. Hassan et al. [198] implemented a multi-sensor PIR system to distinguish fall events, achieving a maximum accuracy of 99% using AdaBoost and 98% using Random Forest with three sensors. The system found to be more effective with multiple sensors and maintained elderly privacy while reducing hardware costs significantly. SiFall proposed by Ji et al. [199] achieved 98.3% accuracy in lab testing and 94.1% real-life detection performance. Over a 3-day continuous trial, the system's false positive rate dropped from 13.4% to 1.4%, showing the model's ability to evolve and adapt to personalized daily behaviors. Arnaoutoglou et al. [193] proposed a low-cost fall detection system using acceleration metrics derived from Doppler signals and a power burst activation mechanism. Their method achieved up to 94.5% accuracy using a shallow neural network and 93.7% with SVM, while a real-time implementation yielded 80% accuracy with zero false alarms across 8 live test scenarios. Pandya et al. [202] applied LBP-based texture analysis on normalized spectrogram, polygram, and tempogram representations of the unknown-2000 and ESC-50 datasets, then trained an LSTM-CNN classifier which achieved an average F1-score of 0.7705 (± 0.2295) and approximately 77% accuracy across varied noise conditions (SNR 0–18dB), consistently outperforming SVM (F10.7259), KNN (0.6832), C4.5 decision trees (0.6526), LSTM (0.7495), and Bi-LSTM (0.7571). The acoustic signal based fall detection system by Lian et al. [203] achieved a 92.6% accuracy and 90.4% recall on fall detection. The system demonstrated robustness with high performance even when deployed in different environments with varied frequencies. Kaur et al. [201] trained an audio transformer using “Diff” features extracted from segmented audio captured during various simulated fall and no-fall scenarios. The system achieved 86.73% accuracy and a 62.34% F1-score, proving effective for detecting verbal or impact-related fall events, though less effective when differentiating distress speech from casual conversation.

Performance Analysis and Design Insights into Ambient Sensor-Based Fall Detection Systems

Ambient fall detection systems offer the advantage of non-contact sensing, making them suitable for privacy-sensitive settings. However, their performance is influenced by environmental factors, sensor coverage, and modality constraints. Threshold-based systems using RFID, infrared, and acoustic sensors [192, 208, 211, 215] are computationally efficient but often lack adaptability, struggling in multi-occupant or noisy conditions. Machine learning methods [193–200, 213] improve generalization and handle diverse signal features but can suffer from data scarcity and hardware variability. Acoustic-based systems [201–205, 218]

offer privacy benefits and can detect falls in occluded environments, but require sophisticated noise reduction and may be impacted by ambient sound interference. Additionally, systems relying on radar, floor sensors, or multimodal fusion [209, 212, 217] show promise but may be limited by deployment cost and calibration complexity. While some systems, such as mmWave- or PIR-based models [197, 198], achieve up to 99% accuracy, others like acoustic or transformer-based approaches [201, 202] yield lower F1-scores, especially under variable noise or speech conditions. Overall, ambient approaches balance privacy and unobtrusiveness, but face challenges in scalability, environmental robustness, and consistent performance across deployment settings.

Performance of Multimodal Fall Detection Approaches

The fall detection performance of different multimodal-based studies is analysed in this section based on datasets and algorithms. Fig. 7 shows the categorisation of different data modalities, popular algorithms applied to them and the

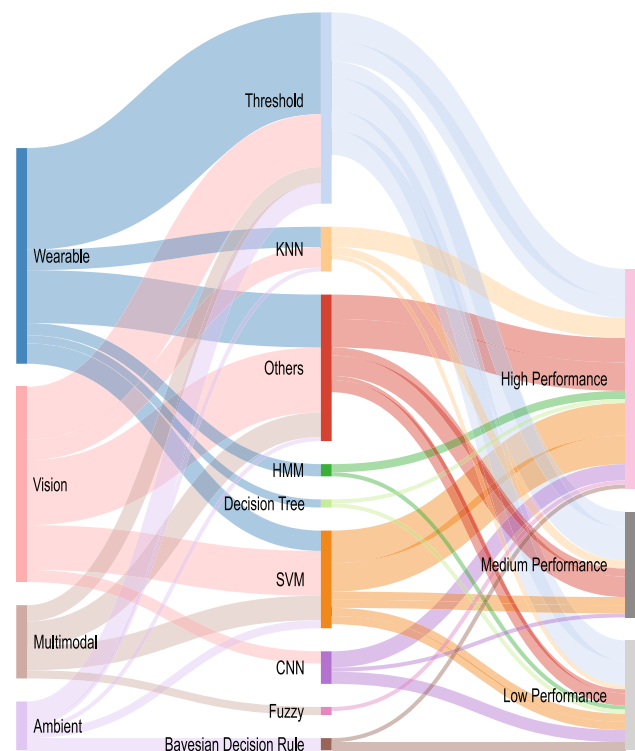


Fig. 7 Categorization of performances of wearable, vision, ambient, and multimodal fall detection approaches based on relative performance. This diagram illustrates the categorization of high-performing, medium-performing, and low-performing methods associated with various AI-based fall detection systems using relative performance metrics. It has been reflected that a high percentage of methods fall in the high-performing category. The approaches and algorithms, as well as their categories, are represented by different colours

performance of these algorithms on those data. A detailed summary of obtained results, datasets and algorithms in multimodal fall detection is presented in Table 15.

Feldwieser et al. [242] carried out statistical analysis to detect elderly fall events employing accelerometer, camera, and microphone. The system detected 2.66 falls per day and successfully detected 12 falls out of 15 actual falls during the study. Falls were analysed using a fall protocol and a medical questionnaire. 28 senior citizens with an average age of 74 participated in the experiment. Tasoulis et al. [245] detected fall events from streaming data collected from the accelerometer and overhead camera. Using a cumulative sum (CUSUM) algorithm, the proposed system could detect fall events with a small number of false alarms and reduced delay. Makhoulouf et al. [223] proposed a multimodal system consisting of accelerometer, photoelectric sensor and oximeter to detect falls, cardiac illness and geo-location respectively. The system achieved a fall detection accuracy of 87%, sensitivity of 82% and specificity of 92%. The data collected from a person can be fused and sent to the doctor.

Zhang et al. [244] employed acceleration, speech recognition and computer vision to detect fall events of a person in home environment and sent alerts to caregiver/medical correspondence. The system can detect falls with 94% accuracy, 3% FPR, and 10% FNR where the response time was 46.2 seconds. Ozcan and Velipasalar [227] combined multimodality sources such as vision sensor and accelerometer sensor to develop a robust portable fall detector and obtained fall detection sensitivity of about 91% and 2% false-positive rate by using histogram edge orientation and GLBP features.

Cippitelli et al. [224] developed a fall detection approach incorporating accelerometer and depth sensor. Using TST Timed Up and Go (TUG) Dataset, the proposed system achieved 99% of sensitivity, 100% of specificity, and 99% of accuracy for detecting falls with high reliability. Kwolek et al. [225] implemented a fuzzy inference engine-based multimodal fall detection system combining an accelerometer and depth sensor which acquired accuracy of 97.14%, precision of 93.75%, sensitivity of 100% and specificity of 95% by using URFD [183] fall dataset. Li et al. [228] incorporated a bi-LSTM model to perceived fall and daily activities with the help of multimodal fusion of accelerometer, magnetometer, gyroscope and FMCW radar data. This model could successfully detect the fall event with an accuracy of 96% using a hybrid fusion approach. Wang et al. [229] developed a fall detection system by taking advantage of multi-sensor fusion and ensemble learning and achieved fall detection accuracy of about 99.30% using an ensemble of CNN classifiers. Moreover, the proposed algorithm detected a fall with a lead time of 350 ms with 0.69% false-positive rate. Ciabattoni et al. [238] proposed a multimodal fall detection system consisting of the RGB camera and audio sensor. The proposed system

Table 15 Dataset and algorithm used in multimodal fall detection approaches

Ref	Algorithm	Dataset	Result
[223]	Threshold	Self	Acc. 87% , location Sen. 94%
[224]		Self	Algorithm 1- Sen. 59% ,Spe. 98% Acc. 79%Algorithm 2-Sen. 79% Spe. 100%Acc. 90% Algorithm 3-Sen. 99%Spe. 100%Acc. 99%
[227]		Self	Sen. : 91% FPR : 0.026
[244]		Self	Acc. : 94% FPR: 3% FNR: 10%
[236]	SVM	Le2i ,SEIN	Le2i-Sen. 90% Spe. 92%; SEIN: 6/9 falls detected, Spe. 99.7%
[237]		N/A	N/A
[239]		Self	Sen. 99% and Spe. 98%
[240]		URFD	Acc. 96.77% Sen. 100% Spe. 100% Pre. 0.9355, Re. 1, F-Score - 0.9667, AUC=0.97
[246]		Self	Acc. 98.2%
[241]		URFD	SVM: Acc. 99.45, Pre. 98.21, Sen. 100.0, Spe. 99.22
[226]	Fuzzy	N/A	N/A
[225]		URFD	Acc. 97.14% Pre. 93.75% Sen. 100% Spe. 95%
[228]	Bi LSTM	Self	Acc. 95%
[229]	Ensemble CNN	Self	Acc. : 99.30% and FPR 0.69%
[238]	CNN-Mobile Net	Self	Sen. 90% Spe. 100% Acc. 93%
[245]	CSUM	self	Reduce false alarm and reduce delay time
[243]	KNN, Kstar, SVM , MLP	Self	Low false alarm
[242]	N/A	Self	12 out of 15 falls were detected
[231]	STCAN+ Bi-LSTM	UP-Fall, UR Fall	UP-Fall: Acc. 99.09%, Pre. 97.06%, Rec. 97.18%, F1-score 96.99%;UR-Fall: Acc. 99.32%, Pre. 99.23%, Rec. 99.19%, F1-score 99.21%
[232]	CNN, ConvLSTM, XGBoost, RF, kNN, LSTM-CNN	UAL-MFDD	CNN on three-camera raw images: F1=0.98
[233]	ResNet, MLP	Le2i, URFD	Multicam: Middle-size multimodal model 98.88% accuracy (+6% vs. single-modal), 98.14% low-light, Frame Masking boosted to 99.63%

Legend: Sen.– Sensitivity; Spe.– Specificity; Acc.– Accuracy; Pre.– Precision; SVM– Support Vector Machine; RF– Random Forest; FPR– False Positive Rate; FNR– False Negative Rate; KNN– K Nearest Neighbour; MLP– Multi Layer Perceptron; CSUM– Cumulative Sum; Pre.– Precision; Re.– Recall; AUC– Area Under The Curve ; CNN– Convolutional Neural Network

employed CNN mobileNet model and detected potential fall event with an accuracy of 93%. Doukas et al. [246] integrated vision and motion sensor data for detecting falls using an SVM classifier. The system correctly classified the fall event with an accuracy of 100% and .0072 RMS error. Zerrouki et al. [240] proposed an EWMA-SVM-based fall detection model combining accelerometer and vision sensor. Using the URFD fall dataset, the proposed EWMA-SVM model outperformed the most commonly used KNN and SVM classifier and achieved an overall accuracy of 96.77%. Tong et al. [239] presented a combination of surveillance camera and smartphones' accelerometer-based fall detection system named as "FallDetect" using SVM. The system detected fall events with an accuracy of above 98%, sensitivity of 99%, specificity of 98%, and outperformed many state-of-the-art models. Geertsema et al. [236] developed a fall detection system for epilepsy patients using SVM and information fusion of video and audio data. This proposed system classified 6 out of 9 real events of falling down with 99.7% specificity using the SEIN dataset and using Le2i dataset, it achieved a sensitivity of 90% and specificity of 92%. Kepski and Kwolek [241] implemented a fall detection scheme embedded with a wearable accelerometer and kinect camera. The proposed model achieved fall detection accuracy of 99.45% using SVM from overhead sequence data of URFD dataset and 95.71% accuracy using KNN from a frontal sequence of URFD dataset.

Shin et al. [231] achieved 99.09% accuracy on the UP-Fall dataset and 99.32% on UR-Fall by fusing motion-aware skeletal features and accelerometer data through a multimodal Bi-LSTM-CA and GSTCAN framework. The model showed high sensitivity (up to 99.19%) and robust performance even in complex fall scenarios, significantly outperforming unimodal approaches and prior state-of-the-art systems. Using leave-one-group-out validation on the custom UAL-MFDD dataset, Lupión et al. [232] showed that a multi-branch CNN on raw thermal images achieved an F1-score of 0.98; IMU-only XGBoost yielded 0.955; and the best fusion of raw+OF from all three cameras reached 0.950. The system also demonstrated 100% binary fall detection accuracy when

requiring two consecutive fall windows, and the Alexa alert workflow reliably filtered false positives via user interaction.

Zhang et al. [233] evaluated Fall-Mamba on a merged Le2i+URFD+Multicam dataset, showing that the middle-size multimodal variant reached 98.88% accuracy under normal lighting and 98.14% accuracy under low-light conditions. Incorporating Frame Masking further boosted accuracy to 99.63% and under 30% mixed noise and occlusion interference, accuracy stayed at 92.62%- ensuring its robustness and resilience in IoT home environments.

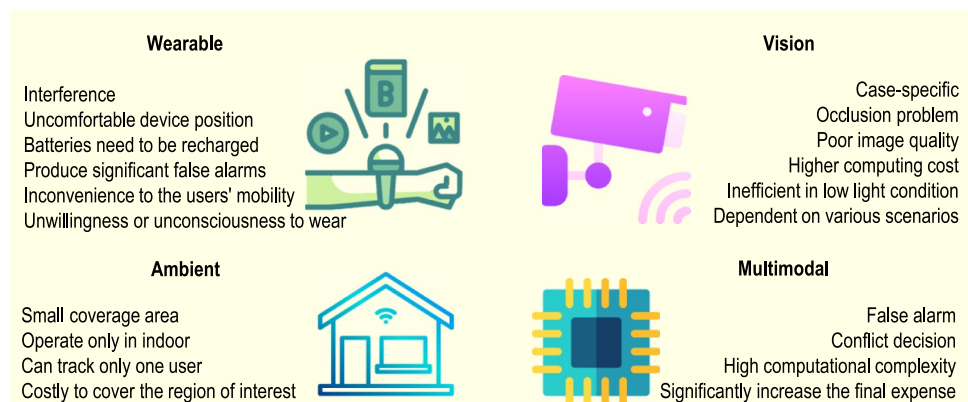
Performance Analysis and Design Insights into Multimodal Fall Detection Systems

Multimodal fall detection systems integrate wearable, vision, and ambient sensors to enhance robustness by leveraging complementary strengths across modalities [223, 226, 228]. Threshold-based fusion methods, such as those by [224, 227], offer low complexity and real-time response but often rely on fixed rules, which limits adaptability in dynamic environments. Machine learning models [229, 231, 233] achieve high accuracy (often over 98%) across various datasets, yet face challenges with synchronization, increased hardware cost, and data heterogeneity. Sensor fusion at the feature or decision level improves generalization [235, 244], but may suffer under missing modality scenarios or sensor failures. Privacy and energy constraints also arise, especially in audio-visual and thermal systems [230, 232], although edge-based and federated frameworks mitigate these issues. Overall, while multimodal FDSs outperform unimodal systems in terms of accuracy and reliability, practical deployment demands trade-offs in complexity, cost, energy efficiency, and privacy preservation.

Existing Challenges and Future Works

There are a number of challenges (see Fig. 8) that need to be addressed which are found as research gaps in the existing works of fall detection methods. This section

Fig. 8 Summarization of challenges and future directions of different sensor-based fall detection systems



outlines the challenges associated with various fall detection methods and provides recommendations for further research.

Wearable devices' significant problems are interference, uncomfortable device position, unwillingness or unconsciousness to wear, and inconvenience to the users' mobility. Unwillingness or unconsciousness to use the gadget may lead to an ineffective wearable system. Unwanted sensors placed on the human body may lead to pain, discomfort, and irritability. Wearable gadgets on the belt around the waist cannot be used for getting dressed or undressed and cannot be carried whenever a user goes to bed. The inclusion of additional sensors contributes to inconvenience and discomfort for the user. An approach to solve these problems is to encourage the users to select the device position. The device can conduct the sensor's localization on the body to detect the device's location on the user, preventing unnecessary placements. Although many falling incidents happen in the shower, the users cannot wear a device in the shower as the device can be hampered by water and user inconvenience. Wearable sensors are battery-powered, which indicates that batteries need to be recharged or changed for proper operation. This issue can be solved by using a low sampling rate and hierarchical system approach. It would also reduce the system's computing complexity, thereby minimising the computational burden. In their operating time, wearable sensors produce significant false alarms, causing user annoyance. The rationale for lower accuracy and higher false positives when using sensors is a lack of adaptability and knowledge of the surroundings. False alarms can be minimised by integrating interaction between the user and the system. Whenever a fall happens, the user will be first alerted to decide if a fall has occurred. Unless the user fails to communicate within a predefined time, the emergency room will be notified.

Vision-based approaches seem to be case-specific as well as dependent on various scenarios. Vision-based algorithms are tested from data obtained from a laboratory setting, experimental conditions including appropriate ambient light, normal circumstances or situations, and falls are emulated by recruited participants. Compared to simulated data, real-world data challenges are that image quality is poor and falling is irregular and differs considerably in terms of velocity and type of fall. Their outcomes differ from scenario to scenario. In the low light condition, vision-based technologies are inefficient since surveillance cameras fail to detect fall events in low light and various lighting states. Another significant drawback is the occlusion problem, although several approaches claim to solve this issue. The vision-based fall detection method often uses complicated computer vision techniques to monitor a person, requiring considerable computational and storage power to run a

real-time automated system with significantly higher computing cost. Participants, while being tracked, need to stay in the camera view. Ethical problems involved with camera-based approaches require the tracked individual's privacy and anonymity, finding it impossible to track the individual in the bathroom and bedroom.

The ambient sensors only operate indoors or when the system is restricted to dead zones, with blind spots and a small coverage space. This system can track only one user. The restricted range of recording would not impact near-field electrical and pressure sensors. However, it would be costly to cover the region of interest. Many environmental systems presume that only one individual is present in the surveillance space. Future works can focus on detecting the fall events of several persons and detecting outside fall events.

A multimodal FDS incorporating more than one mode of sensing elements improves the system's stability. However, adding more than one sensing mode to the surveillance system may significantly increase the final expense. Different sensors may treat similar events differently during the monitoring period, leading to false alarms. False alarms also occur with some conventional fall detection systems. The data analysis process for most multimodal systems involves evaluating data for each element individually prior to leveraging fusion algorithms to aggregate them. This causes the fusion algorithm to be excessively complicated, raising computational complexity. As the number of monitoring components increases, the more data the system captures, and the more computational costs the system incurs.

One of the critical factors affecting fall detection research is that of simulated and real fall data. Simulated data offers a controlled environment for maintaining consistency and repeatability; however, it does not possess the variability or unpredictability that might occur in real life. Real fall data has invaluable advantages during the validation of performance, but their collection poses quite a few problems, starting with ethical ones, up to the difficulty of recording authentic falls. Future efforts should prioritize incorporating real data into analyses while leveraging simulated data for initial model development, ensuring both reliability and generalizability in practical applications.

Another critical yet under-discussed issue in fall detection datasets is subject and environment diversity. Many existing datasets include a narrow range of participants—often healthy, young adults—falling in controlled environments, which limits the model's ability to generalize to real-world settings involving elderly individuals, various body types, or occluded environments. This bias impacts performance, particularly when systems are deployed across populations not represented during training. Future research should incorporate fairness-aware training protocols, diverse data

acquisition strategies, and domain generalization techniques to enhance robustness and equitable performance across user groups.

An important challenge in the research on fall detection is related to methodologies employed in validation. The vast majority of works are based on internal cross-validation, which relies on random splits of available data into training and test sets. While it does evaluate model performance, it mostly lacks the generalization capability to new, previously unseen scenarios like different users or different environments. External validation techniques, such as leave-one-subject-out cross-validation, could give a more realistic estimation of performance by training a model on completely new subjects to simulate real-world scenarios. Future research is encouraged to adopt external validation techniques for better assessment of the strength and generalization of fall detection systems under varying settings and populations.

Real-time fall detection technologies are highly relevant to ensure immediate intervention and safety for the user. Systems using this technology must resolve issues of latency minimization, computational efficiency optimization, and feasibility of implementation in diverse settings. Although a few solutions already depict their potential, most of them suffer from the trade-off between speed and accuracy, particularly in resource-constrained devices. Future research must be directed to improve real-time algorithms with a reduction in system complexity, ensuring robust performance in dynamic settings, which would make fall detection scalable and effective.

Conclusion

Elderly fall is the second leading cause of death among all injuries. It causes great effects on the elderly independent living as there is no one to rescue the older adults during falls immediately. Many studies have been developed to date to detect and predict fall incidents at the earliest time. This study presented the existing fall detection approaches based on artificial intelligence algorithms as well as the sensors they have utilised. A systematic literature search has been performed in popular databases such as Scopus, PubMed, Google Scholar, IEEE Xplore. More than 145 research articles have been chosen carefully as per PRISMA guidelines. We have categorised the fall detection approaches into four different types, including wearable sensors, vision sensors, ambient sensors, and multi-modal sensor-based systems. Furthermore, we have subdivided each fall detection approach into different categories based on the methods they used, including threshold-based algorithms and different AI algorithms. The publicly available datasets were discussed.

The study was designed to give researchers a clear and thorough idea about AI-based fall detection approaches. Moreover, this paper highlights a detailed description of the current research challenges that need to be addressed in the future.

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Author Contributions This work was carried out in close collaboration between all co-authors. First defined the research theme and contributed an early design of the system. Further implemented and refined the system development. Wrote the paper. All authors have contributed to, seen, and approved the final manuscript.

Data Availability We do not analyse or generate any datasets because the current work proceeds within the existing literature.

Declarations

Conflicts of Interest Statement The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Ethical Approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed Consent Informed consent was obtained from all individual participants included in the study.

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